Deep research report

Data, open-source projects, current technology and integration paths for the AquaSync dam–river digital twin

Produced by a 13-agent research sweep across six domains.
Every URL curl-verified; every claim checked by a second, adversarial agent.

144	30	14	561.0 MB
sources confirmed after verification	claims rejected as unverifiable	resources downloaded into research/	acquired and indexed locally

Why every link in this document was verified by curl.

The planning material this project grew out of was full of confident, wrong references: one Amazon ASIN appeared as five different products, and a dataset was described as containing 2018 flood data it does not contain. Both errors survived because nobody checked.

So each research agent was required to run `curl -o /dev/null -w "%{http_code}"` against every URL and record the real status, and a second agent then re-checked the list adversarially and rejected what it could not confirm. 30 claimed sources did not survive that pass.

Generated 27 August 2026 · Regenerate with `scripts/build_research_report.py` *· Local index at* `research/index/README.md`

1 · Executive summary

The findings that change what AquaSync should do next.

1. CONSTANTS BUG, verified twice: d:\aqua_sync\backend\aquasync\twin\constants.py sets Idukki mwl=732.43 (identical to FRL) and Idamalayar mwl=169.00. CAG Appendix 3.1 (I extracted it from research/sources/papers/CAG_Kerala_flood_audit.pdf, page index 119) gives Idukki MWL 734.11 m / Idamalayar 171.20 m, and the live KSEB July-2026 workbook independently lists '2408.5 ft' (733.91 m) and 171 m. AquaSync therefore models ZERO surcharge above FRL. Integrating the RAT area-elevation curve over 732.43->734.11 m gives ~90 MCM of storage the optimiser cannot see - comparable to the entire ~90 MCM flood cushion CWC says operators actually had in August 2018.

2. SPILLWAY BUG: backend/aquasync/twin/reservoir.py line 206 sets `crest = self.res.red_level` (728.19 m for Idukki). The published Idukki spillway crest is 2373 ft = 723.29 m (KSEB July-2026 workbook 'Spillway Crest Level' column, which I downloaded and parsed; also a KSDMA bulletin column). At FRL the current weir formula returns ~1,400 cumec; with the true crest it returns ~4,475 cumec against the official design discharge of 5,012 cumec (CAG App 3.1). Gate geometry (5 radial gates x 12.19 m) is correct. Every 'we could have released more, earlier' conclusion is currently capped by a wrong constant that understates Idukki's discharge capacity roughly 3x.

3. THE RULE LEVEL IS A CURVE, NOT A SCALAR, AND AQUASYNC FROZE IT AT THE 31-AUGUST VALUE. I read all 36 values of CAG Appendix 3.3 (the KSEBL 2020 rule curve, 18 ten-day steps, both dams). AquaSync's rule_level=728.50 (Idukki) and 164.00 (Idamalayar) are exactly the 31-August rows. I then proved the curve is live and applied as a STEP function holding the end-of-period value: the KSEB July-2026 workbook carries rule 2375.33 ft on BOTH 01 and 10 July (=724.00 m = CAG's 'July 10th' row), 2377.95 ft on 20 July (=724.80 m), 2380.58 ft on 31 July (=725.60 m) - an exact match to CAG - and CAG records KSEBL's own reply that the 11-20 August level is 2,386.81 ft (=727.50 m = CAG's 'August 20th' row). This matters because optimizer.py line 229 normalises headroom by (frl - rule_level) and line 485 searches targets from blue_level upward.

4. IDAMALAYAR ALERT BANDS ARE WRONG BY 1.0 m AND 0.5 m. Alert levels are a fixed offset below the rule level: Idukki Blue/Orange/Red = rule minus 8.00/2.00/1.00 ft (AquaSync's Idukki offsets are correct); Idamalayar = rule minus 1.50/1.00/0.50 m, which I verified holds at all four sheets I parsed in the July-2026 workbook (rule 161.5 -> 160/160.5/161; rule 162.5 -> 161/161.5/162) and again in the KSDMA 25-08-2026 bulletin (rule 164.00 -> 162.50/163.00/163.50). AquaSync has Idamalayar blue=161.50 (rule-2.50) and orange=162.50 (rule-1.50). Blue fires 1.0 m late, orange 0.5 m late; red is right.

5. A GENUINE DAILY OBSERVED 2018 INFLOW SERIES FOR BOTH RESERVOIRS IS ALREADY IN THE REPO, AND I PROVED WHICH COLUMNS ARE REAL. CAG Appendices 3.6 and 3.7 give 112 daily rows each, 10-06-2018 to 29-09-2018. I diffed App 3.5 against App 3.6 across their 92 common dates: the inflow column is identical 92/92, while reservoir level is identical 0/92 and spill only 38/92 - so inflow is observed forcing and level/spill are counterfactual simulation. Cross-validation is excellent: CAG's Idukki 15-17 Aug inflow sums to 431.7 MCM against CWC's independently stated 435 MCM, and CAG's Idamalayar peak day of 100.59 MCM = 1,164 cumec matches CWC's stated 1,164 cumec exactly. Gap #1 is closed for the forcing a reservoir twin actually needs, with zero downloads.

6. THE PERFECT-FORESIGHT GAP NOW HAS A TRAINING SET. I broke open the Kerala SLDC historical form: POST to https://sldckerala.com/index.php?id=7 with body date1=YYYY-MM%2FDD-YY&date1;_day=&date1;_month=&date1;_year=&sbtstore;=SHOW returns daily OBSERVED inflow (mcm/day), spill, rainfall, level, storage and cumulative inflow for both reservoirs. Confirmed live: 2021-10-20 -> Idukki 730.959 m / spill 9.072 / inflow 29.793; 2020-09-25 -> 1209.408 MCM / 83%; 2021-04-30 -> 524.654 MCM / 36%; 2019-08-08 -> inflow 135.532 MCM/day with 194.2 mm rain. 2018-08-16 returns an empty shell, so the archive genuinely starts 2019-08-08 (~2,575 days to today). It also independently repairs the ENTIRE 2020-09-25 to 2021-04-30 window that is corrupt in the amith-vp feed.

7. THE RAT-KERALA IDUKKI AREA-ELEVATION CURVE IS PHYSICALLY BROKEN - DO NOT USE IT FOR STORAGE. I downloaded and integrated it: surface area jumps from 3.005 km2 at 722 m to 43.1 km2 at 723 m (a 40.1 km2 step in one metre), and the MDDL->FRL integral gives 532 MCM against the official live storage of 1,459.49 MCM - 2.7x too small. Areas above ~724 m are plausible (50.015 km2 at 730 m vs Suresh et al.'s 60 km2 at FRL) and are fine for evaporation, rain-on-lake and the surcharge estimate. Prior research advised cross-checking AquaSync's r2=0.9957 level-storage law against this curve; doing so naively would have injected a 2.7x error.

8. THE PUBLISHED RULE CURVE WOULD HAVE MADE IDUKKI WORSE, SO 'COMPLY WITH THE RULE CURVE' CANNOT BE THE HEADLINE. CAG Table 3.6/3.7 (read directly): actual Idukki spills 14-18 Aug = 467.51 MCM; the 2020 rule curve gives 531.03 MCM; the 1983 curve started from the upper or actual level gives 558.19 MCM; only a 1983 curve initialised from the LOWER rule level gives 308.13 MCM. Idamalayar's 2020 curve does help marginally (323.49 vs 340.50 actual). CAG also records that KSEBL told the IISc team no rule curve was followed at all during the flood despite possessing the 1983 curve, and that a 'dynamic flood cushion of four feet below FRL (68.87 MCM)' was introduced in August 2018 and still did not avoid 467 MCM of spill.

9. ROUTING IS ROUGHLY 2x TOO SLOW AGAINST THE ONLY OFFICIAL ANCHOR. AquaSync's reaches give Idukki->Aluva K = 11 + 8 = 19 h. CWC's December-2018 report states an 8-hour travel time for combined Idukki + Idamalayar dam discharges to Neeleswaram (which sits between AquaSync's Perumbavoor and Aluva control points). Also fatal to a previously proposed calibration route: ARANGALI is on the Chalakudy, not the Periyar - the NWDP file's own Tributary/Local River/District columns say Chalakudy/Chalakudy/THRISSUR - so Neeleeswaram+Arangali is NOT an inflow-outflow pair. The defensible reach is dam outflow -> Neeleeswaram, with Arangali repurposed as the Chalakudy lateral-inflow boundary below it.

10. INTEGRATION IS A READ-ONLY PUBLISHING PROBLEM WITH A STATUTORY HOOK AND A NAMED DECISION FORUM - NOT A SCADA PROBLEM. Dam Safety Act 2021 s.35(1)(a) makes an inflow forecasting system a dam owner's statutory duty and s.35(1)(e) requires anticipated inflows/outflows in the public domain; s.41/s.42 attach personal criminal liability (up to 2 years, head of department deemed guilty), which explains operator conservatism better than any technical barrier. The actual decision point is Kerala GO(Rt) No. 453/2021/DMD - a multi-agency committee that convenes when a dam reaches ORANGE alert. Idukki's instrumentation is Encardio's closed RESHMI/Drishti stack under DRIP, and CEA's 2021 guidelines state as a cardinal principle 'hard isolation of their OT Systems from any internet facing IT system'. There is no evidence of any Modbus/IEC-104 endpoint to talk to; do not plan a field-bus integration.

2 · Evidence that complicates the thesis

The most useful part of any research sweep is what it finds against the hypothesis. These are the points a domain-literate reviewer will raise, and it is far better to have raised them first.

Challenge 1

THE LITERATURE DOES NOT AGREE ON WHETHER DAM OPERATION WORSENED THE 2018 FLOOD, AND BOTH CAMPS ARE CREDENTIALED. CWC's own September-2018 study concludes the dams attenuated the peak (Idukki 2532 -> 1500 cumec, 'an attenuation of about 41%', FRL touched 01:00 on 17 August) and states explicitly that even a 75%-filled reservoir could not have mitigated the event. Sudheer et al. (Current Science, IIT Madras) agree the dams helped: a virgin no-dam simulation peaks at 11,990 cumec at Neeleswaram against the observed ~8,800, and only 16-21% peak attenuation was ever achievable by advance emptying. Against them stands CAG/IISc's official performance audit, which finds that following EITHER published rule curve would have produced MORE Idukki spill (531.03 MCM under the 2020 curve, 558.19 under the 1983 curve from the actual level, versus 467.51 actual) - and which records KSEBL telling the IISc team that no rule curve was followed at all. AquaSync must present this as a live controversy, not a settled result.

Challenge 2

THE MOST-QUOTED 'RESERVOIRS MADE IT WORSE' STUDY WAS REJECTED, AND ITS HEADLINE NUMBER CONTRADICTS THE PEER-REVIEWED ONE BY 3.5x. Mishra et al.'s HESS preprint (hess-2018-480) carries the status line 'The manuscript was not accepted for further review after discussion.' It reports upstream rainfall return periods of more than 500 years; the peer-reviewed Current Science paper reports 145 years for the Periyar basin. Do not put the 500-year figure in any AquaSync-facing material. The preprint's genuinely useful content is its data trail - seven Kerala reservoirs' storage from India-WRIS for 2007-2018, with Idukki at 92.5% of FRL on 8 August 2018, which independently agrees with the CWC bulletin's 92% for 01-08-2018.

Challenge 3

FIRO TOOK ROUGHLY A DECADE OF SHADOW OPERATION TO CHANGE A MANUAL THAT HAD BEEN TOUCHED TWICE IN 66 YEARS. CW3E's steering committee formed in 2014. The Lake Mendocino Final Viability Assessment (December 2020) only RECOMMENDED a five-year major deviation plus a water control manual update - at that point FIRO ran through deviations, not through the manual. A NEPA Environmental Assessment carried it from WY2021. The revised manual was signed on 22 October 2025 and authorises exactly the same 11,650 acre-feet the deviation had already allowed. Scripps' own words: it 'redefines the operating rules around flood control schedules for the first time in the dam's 66-year history' (original 1959, updated 1986, minor revisions 2003). AquaSync should plan for advisory shadow operation measured in years and should say so up front rather than implying deployment.

Challenge 4

THE FORECAST SKILL FIRO ASSUMES DOES NOT EXIST IN THIS BASIN. Sudheer et al. (Current Science 116(5) p.789, verified verbatim): 'The probability of detection (POD) of higher rainfall (>25 mm/day) at short range (3-day) or medium range (5-day) even with the ensemble of models is still low at about 20% to 30% only with a large probability of false alarm.' Lake Mendocino's verified CNRFC numbers are 24-hr inflow R2 of 0.84/0.80/0.73/0.66/0.53 at days 1-5. Mishra et al. concluded operations need skilful extreme-rainfall forecasts at 4-7 day lead. CWC's own December-2018 report says flatly 'There is no Flood Forecasting network in Kerala till date', and I could not confirm from CWC's AFF page that the Periyar or Neeleswaram is covered at all. One honesty note on the citation: Sudheer et al. are quoting their own references 23-24 here, not reporting an original result.

Challenge 5

EVEN A PERFECT OPTIMISER HAS A SMALL CEILING HERE, AND THE BOUND COMES FROM THE PRO-DAM SIDE. Sudheer et al. put achievable peak attenuation from advance emptying at 16-21%, and find that below 50% storage downstream flows become insensitive to reservoir state at all. Bank-full is about 3,400 cumec upstream and 4,200 at Neeleswaram against an 8,800 cumec event. CWC computes that of the ~8,800 cumec at Neeleswaram, about 7,300 came from the free catchment DOWNSTREAM of the dams against roughly 1,500 cumec of Idukki release. CAG Table 3.4 puts the two dams' combined contribution to Neeleswaram flow at 46.43% on 14 August falling to 29.54% on the 16th and 16.99% by the 18th. A twin that optimises the dams is optimising a shrinking minority of the flood, and the honest framing is marginal improvement plus better warning, not flood prevention.

Challenge 6

THE OFFICIAL RECORD CONTRADICTS ITSELF ON THE QUANTITIES AQUASYNC WOULD VALIDATE AGAINST. CAG Table 3.4 reports observed Neeleswaram flow of 793.93 MCM on 16-08-2018, which is a 9,189 cumec daily mean - and that EXCEEDS the 8,800 cumec instantaneous peak the CWC report attributes to the same day. A daily mean cannot exceed the instantaneous peak, so at least one official figure is wrong, or the 7AM-7AM accounting window and the rating curves differ materially between IISc and CWC. Separately, CWC's September-2018 report gives the Periyar area to Neeleswaram as 4,033 km2 in the text, 4,035 in the rainfall table and 4,076 in the catchment table; its December-2018 report gives the Neeleswaram intervening free catchment as 2,329 km2 and then 3,042 km2 two sentences later, for the same unit hydrograph. Never quote these to a precision the sources do not support.

Challenge 7

THE FLOOD PEAK WAS NEVER GAUGED AT THE SITE AQUASYNC ROUTES TO. NWDP's CWC daily discharge for NEELEESWARAM is missing 2018-08-16 through 2018-08-22 (and 24-27): the last pre-gap reading is 6,166.426 m3/s on 15 August and the next is 924.282 on 23 August. The all-time public daily maximum for that gauge is 6,323.5474 on 2013-08-05 - BELOW the 8,800 cumec attributed to 16 August 2018. So the headline 8,800 figure is report-derived, not gauge-derived, and its 14:00/15:00 timing is read off Figure 3 rather than stated in the text. VANDIPERIYAR did record all 31 days of August 2018, but it sits upstream of Mullaperiyar's inter-basin diversion, Idukki and Idamalayar, so it cannot serve as an unregulated upstream partner for Neeleeswaram. Note also a coordinate conflict between two of the source datasets: GUARDIAN places VANDIPERIYAR at (9.1525, 77.1525) in Tamil Nadu while NWDP places it at (9.57333, 77.09056) in Kerala, ~47 km apart; Neeleeswaram and Arangali agree exactly, isolating Vandiperiyar as the bad record. Use the NWDP coordinates.

Challenge 8

AQUASYNC STILL WILL NOT HAVE GAUGE-CALIBRATED ROUTING AFTER THIS WORK. The best anchor available is CWC's 8-hour travel time - and CWC states it was derived 'only for 2018 using the outflow hydrograph from calibrated Mike-11 model at Idukki and Idamalayar dams' precisely because 'hourly discharge from these project authorities were not available'. It is modelled outflow, one event, one basin. Compounding this, CWC records that Idukki has spilled on only three occasions in its entire history - 1988, 1992 and 2018 - so there are effectively three events in the whole record on which a dam-release routing calibration could ever be based. Gap #2 is narrowed to a defensible government prior; anyone claiming it is closed is overstating.

Challenge 9

THE METHOD AQUASYNC USES IS UNDER PUBLISHED ATTACK BY A SERVING CWC DIRECTOR, AND THE JOURNAL REFUSED TO PRINT THE REBUTTAL. J. Harsha (Director, CWC Chennai) argues that SCS-CN with lambda=0.2 came from USDA watershed data 'developed based on limited set of watershed data without peer-review', was never developed for the Periyar, and that using 2010 land cover for a 2018 event 'assumed stationarity of runoff processes in PRB'. Current Science kept the rebuttal for six to seven months and then refused it without peer review, so the exchange lives on SANDRP across three separate posts. AquaSync uses exactly SCS-CN with an antecedent-moisture shift, so it inherits the critique wholesale. Carry the defence explicitly (Sudheer et al.'s published argument that C is insensitive to lambda when CN and P are large, which is the high-flow wet-AMC case AquaSync models) and carry the concession too - the original authors granted 16.51% change in built-up area between 2010 and 2016, and CAG documents Kerala built-up area growing 450% from 60 to 330 km2 between 1985 and 2015 while water bodies fell 17%.

Challenge 10

THE NEAREST PUBLISHED PRIOR ART TO AQUASYNC'S THESIS IS PAYWALLED, AND AQUASYNC CANNOT CURRENTLY SAY HOW IT DIFFERS. Das et al. (2025), 'Forecast-Informed Reservoir Operations within a Satellite-Based Framework for Mountainous and High-Precipitation Regions: Case of the 2018 Kerala Floods', Journal of Hydrologic Engineering 30(2), doi 10.1061/JHYEFF.HEENG-6276 - same UW/SASWE group that published the RSE paper, same basin, same event, same core idea. OpenAlex reports is_oa=false and there is no green copy anywhere; ASCE blocks automated access entirely. Any novelty claim AquaSync makes before someone reads this paper is unsupported. Related caution on the satellite route it uses: Suresh et al. report average cloud cover above 40% over Kerala's dams for 2017-2019 and 'over 80%' during the monsoon, concluding such monitoring is 'not possible without the use of cloud-piercing SAR imagery' - which is exactly the season AquaSync cares about.

3 · Prioritised upgrades

What to build next, each tied to a specific verified project or dataset. Ordered by verdict, not by how interesting it sounds.

	Upgrade	Builds on	Effort	Why it is worth it
do-now	**Correct the reservoir constants: Idukki MWL 732.43 -> 734.11 m, Idamalayar 169.00 -> 171.20 m; spillway crest in reservoir.py from red_level to the published 723.29 m (Idukki) / 161.00 m (Idamalayar); Idamalayar blue 161.50 -> 162.50 and orange 162.50 -> 163.00.** closes: None of the five named gaps - but it invalidates current optimiser output, so nothing else is trustworthy…	CAG Appendix 3.1 in research/sources/papers/CAG_Kerala_flood_audit.pdf (pdf page index 119; I extracted it: Idukki FRL…	0.5 day.	The twin currently believes Idukki can pass ~1,400 cumec at FRL when it is rated at 5,012, and that it has zero storage above…
do-now	**Replace the scalar rule_level with the 18-step CAG 2020 rule curve implemented as a STEP function holding the end-of-period value, and derive blue/orange/red from it by fixed offset (Idukki rule minus 8.00/2.00/1.00 ft; Idamalayar rule minus 1.50/1.00/0.50 m).** closes: Policy realism - optimizer.py normalises headroom by (frl - rule_level) and searches targets upward from…	CAG Appendix 3.3 (all 36 values verified from the local PDF), validated against three sheets I parsed in the KSEB…	1 day.	AquaSync currently targets the 31-August level all year. Worse, a naive linear interpolation of the published curve would sit…
do-now	**Extract the CAG 2018 daily inflow series for both reservoirs and stand up August 2018 as a second validated case study alongside October 2021.** closes: Gap #1 (no genuine pre-2020 operational data) - for the forcing, which is what a reservoir twin actually…	CAG Appendices 3.6 and 3.7 in the local PDF: 112 daily rows each, 10-06-2018 to 29-09-2018. I proved the inflow column…	2 days - PyMuPDF text extraction. The CAG tables extract cleanly (I did it), unlike the Current Science and NTRS paper tables which come out with shifted columns.	Turns the flagship from one case study into two, with the second being the event every reviewer and every KSEB engineer will ask…
do-now	**Re-run the existing policy search under degraded and ensemble inflow instead of the oracle, and publish the loss curve as a headline result.** closes: Gap #3, perfect foresight - the project's stated biggest weakness.	SLDC daily observed inflow 2019-08-08 to present as the historical sample; wltaylor/Robust-FIRO (MIT, 194-line…	1-2 weeks.	Converts the biggest weakness into the most defensible result: 'here is what this policy is worth at the 20-30% POD that is the…
do-now	**Build the SLDC daily archive scraper; use it to repair the corrupt window and as the observed-inflow training and validation set.** closes: The known 11% corruption (2020-09-25 to 2021-04-30) outright, plus ~12 months of history before the amith-vp…	https://sldckerala.com/index.php?id=7 by POST - I verified it live at 2019-08-08, 2020-09-25, 2021-04-30 and…	2 days.	~2,575 days of observed daily inflow, spill, rainfall and level for exactly AquaSync's two reservoirs, from a state authority…
do-now	**Add a deviation-from-published-schedule penalty to the objective function, distinct from the existing ramp-smoothness term.** closes: Adoption, not a technical gap - and adoption is what the project is ultimately for.	Koo, Abraham, Jonoski & Solomatine (2025), arXiv 2407.04506 / J. Hydroinformatics doi 10.2166/hydro.2025.019, verified…	0.5 day - one term in ObjectiveWeights and one in evaluate().	AquaSync already penalises ramp magnitude via gate_movement, but not deviation from the schedule the operator has already…

	Upgrade	Builds on	Effort	Why it is worth it
do-next	**Add the Lower Periyar and Bhoothathankettu nodes to the reach network from CAG Appendix 3.1.** closes: Cascade co-optimisation; also improves lateral-inflow area scaling in the routing.	CAG Appendix 3.1, which I read directly: Lower Periyar dam (KSEBL) FRL 253.00 / MWL 256.00 / MDDL 237.74 / gross 5.30…	3 days.	Lower Periyar has negligible storage (4.5 MCM) but a real 339-cumec powerhouse diversion that AquaSync currently ignores.…
later	**REJECT (for now): 2D inundation modelling.** closes: Gap #4 - genuinely, but it is the least valuable of the five right now.	If and when it proceeds: Deltares/SFINCS (GPL-3.0 source, 83 stars, pushed 2026-08-24) + hydromt_sfincs + pyflwdir…	2-4 weeks minimum, plus an unresolved DEM licence decision.	Real, but it cannot touch the biggest gap: routing and inundation tools cannot fix a forecasting problem. The licence trade is…
reject	**REJECT: using the RAT-Kerala Idukki area-elevation curve as a check on, or replacement for, AquaSync's level-storage law.** closes: Nothing - and prior research recommended it, which is why it needs an explicit reject.	http://210.212.226.241:8080/RAT_Kerala_CWRDM/assets/RAT_results/aec/idukki.csv, which I downloaded and integrated.…	n/a.	Adopting it would inject a 2.7x storage error into a model currently calibrated to r2=0.9957. Salvage the part that is sound:…
do-now,	**Recalibrate Muskingum K against the CWC 8-hour published travel time and the CAG observed downstream daily volumes, and reframe the calibration reach as dam-outflow -> Neeleeswaram rather than gauge-to-gauge.** closes: Gap #2 (uncalibrated K and x) - narrows it substantially; does not close it.	CWC December-2018 report sec 8.3.2.2 (verified verbatim: level forecast for Neeleswaram from Idukki and Idamalayar dam…	1 week.	AquaSync's Idukki->Aluva K of 19 h is roughly 2.4x CWC's 8 h to Neeleeswaram, which shifts every arrival-time statement the twin…
do-next.	**Add Mullaperiyar as an exogenous upstream boundary so Idukki inflow stops being treated as natural.** closes: Cascade co-optimisation (partially) - and it is the largest single unmodelled control in the basin.	RAT-Kerala inflow/mullaperiyar.csv (verified 200, 100,535 bytes) for a daily series; Suresh et al. 2024 attribute…	1 week.	A mass-balance twin that calls Idukki inflow 'natural' is structurally wrong: Mullaperiyar intercepts and diverts flow…
do-next.	**Force the SCS-CN module with IMD 0.25-degree gridded rainfall instead of gauge rainfall, and write the lambda=0.2 defence into the docs.** closes: Gap #1 (pre-2020 forcing) plus a live, published methodological attack on exactly AquaSync's method.	IMD POST endpoint https://www.imdpune.gov.in/cmpg/Griddata/RF25.php with body RF25=YYYY - I ran it live for 2018: HTTP…	3 days.	The Kerala gauge rainfall file has effectively NO long record inside the Idukki catchment - filtering to the true headwater box…

*A **reject** verdict is a result, not a gap. The failure mode this project is most exposed to is feature creep across twenty-five plausible upgrades, and a list that never says no is no help against it.*

4 · Integration paths

How AquaSync would reach the systems KSEB, KSDMA and CWC already run. The technical routes are mostly solved problems; the blockers are institutional, and are recorded as such.

KSDMA multi-agency dam operations committee (the real decision forum)

Approach. AquaSync produces a pre-computed options brief, not a control action: 3-5 candidate release schedules with peak level, peak downstream discharge at Aluva, spill volume, forgone generation and time-to-FRL, delivered when a dam crosses orange alert. Mirror the KSDMA bulletin schema exactly (FRL, rule level, blue/orange/red, live storage at FRL, live storage, % storage, current spillway release in cumecs) so the committee reads familiar columns.

Protocol or standard. No protocol - a one-page PDF/HTML brief matching the KSDMA daily bulletin layout. Optionally OGC API - EDR (OGC 19-086r6) later for machine access.
Effort. Low - days, once the rule-curve and constants fixes land.
Blockers. Institutional, and larger than technical. The committee decides, the operator executes; AquaSync has no seat. Kerala's rule curves are described as 'prepared for all major dams' but are NOT published: on the KSDMA Dam Management page only Kallarkutty carries a link among 19 KSEB dams and the Idukki-Cheruthoni EAP cell is empty. Obtaining the authoritative current Idukki rule curve is a written request, not a download - the CAG Appendix 3.3 copy is what you have until then.

Regulatory. G.O.(Rt) No. 453/2021/DMD dated 3-6-2021 constitutes the committee and states verbatim that it 'meets when dams reach their orange level of alert or if extreme rainfall events are forecasted'. Verified at https://sdma.kerala.gov.in/dam-management/ (200).

KSEB Ltd Dam Safety Organisation - historical bulletin archive

Approach. Ingest the monthly-statistics workbooks as the authoritative KSEB-origin history: 86 downloadable files, June 2019 (wpdmdl=315) through July 2026 (wpdmdl=6380). Modern sheets carry Rule level, Blue/Orange/Red, Live Storage, Inflow (MCM), Average Inflow (Cumec), Power House Discharge, Spill (MCM) and Current Spillway release - i.e. observed release partitioning AquaSync's power and spill models can be calibrated against.

Protocol or standard. None - WordPress Download Manager IDs serving .xlsx (to Oct 2020) then legacy .xls (Nov 2020 onward).
Effort. Low - 1-2 days for a loader.
Blockers. IDs 310-314 (Jan-May 2019) are dead: they 302-redirect to kseboa.in which does not resolve, so the archive starts at June 2019, not January. The format flips to legacy .xls from ID 1178 (Nov 2020) - openpyxl refuses these and that stretch is exactly the corrupt-window months you most want, so budget xlrd or a LibreOffice conversion step. The 2019 workbooks' 'Previous Year Water Level' column yields only 96 unique 2018 dates, of which just three are in August (20th, 30th, 31st) - the flood peak is not there.

Regulatory. Site content is CC BY-SA 2.5 IN. KSEB Ltd is the specified-dam owner under Dam Safety Act 2021 and the counterparty for any formal data agreement.

Kerala State Load Despatch Centre (SLDC) - daily observed inflow archive

Approach. Scrape the date-queried System Statistics: Storage table for both reservoirs, daily from 2019-08-08 to present. Yields level, storage, % storage, rainfall (mm), spill, INFLOW (mcm/day) and cumulative inflow - the single best free training set for a Periyar inflow forecast model.

Protocol or standard. None - HTTP POST to a PHP form. Body: date1=YYYY-MM%2FDD-YY&date1;_dp=1&date1;_day=DD&date1;_month=MM&date1;_year=YYYY&sbtstore;=SHOW. GET does not work; the two-digit year suffix in date1 is required.
Effort. Low - 2 days including a polite rate-limited crawler and a parsed CSV cache.
Blockers. Undocumented endpoint with no rate-limit statement or terms - one request per day of history, so crawl slowly and cache. Archive does not reach 2018 (2018-08-16 returns an empty shell; 2019-08-05 empty, 2019-08-08 populated). Critically, the three official Kerala sources DISAGREE on the same date: for 25-08-2026 the KSDMA bulletin gives Idukki 2364.22 ft with bands 2382.09/2388.09/2389.09, while the KSEB dashboard and SLDC give 720.968 m (~2365.38 ft) with bands 2389.78/2395.78/2396.78 - two alert-band sets ~7.7 ft apart. Reconcile explicitly before treating any as a hard constraint.

Regulatory. None specific; SLDC operates under the Electricity Act / CEA framework, so any formal feed request travels through KSEB rather than SLDC directly.

Central Water Commission - inflow forecasting (the route out of perfect foresight)

Approach. Request Periyar/Neeleswaram inclusion in CWC's inflow-forecast programme and access to the Advisory Flood Forecast product, then feed its ensemble into AquaSync's policy search in place of the oracle inflow array. CWC's page confirms 'Inflow Forecasts at 128 reservoirs are used by the dam authorities in timely operation of reservoir gates'; AFF runs 20 basin models every three hours with a 7-day horizon on satellite rainfall, rain gauges, IMD GFS and WRF.

Protocol or standard. No open API. AFF is a web product; ingestion would be a negotiated feed or a scrape of a station page.
Effort. Medium-to-high - this is a letter and a relationship, not a download. Months.
Blockers. PERIYAR COVERAGE IS UNPROVEN. The AFF landing HTML contains no occurrence of 'Neeleswaram', 'Periyar' or 'Kerala' (stations load via JavaScript), and CWC's own December-2018 report states flatly 'There is no Flood Forecasting network in Kerala till date'. Verify interactively before planning around it. India-WRIS, the other national route, is unreachable: DNS resolves 164.100.85.36 but TCP to :443 times out, so the recovered API contract is documentation, not access.

Regulatory. Dam Safety Act 2021 s.35(1)(a) obliges the dam owner to 'establish well designed hydro-meteorological network and an inflow forecasting system', and s.35(1)(f) obliges assistance to the Authority for 'exchange of real time hydrological and meteorological data'. That is the lever for the request.

NDSA / State Dam Safety Organisation - statutory reporting

Approach. Make AquaSync emit the SDSO-facing instrument-reading analysis as a first-class output, so the tool is a compliance deliverable rather than an optional extra. Poll the open NDSA JSON API for newly gazetted regulations rather than re-reading a page: GET /ndsa-backend/web/api/document-config, then POST /ndsa-backend/web/regulation?lang=en with {"page":0,"sort":"","search":""}.

Protocol or standard. Undocumented but open JSON API (no key). Content governed by NDSA regulations under the Dam Safety Act 2021.
Effort. Medium - low to poll the API, medium to build a report once the form is known.
Blockers. The regulations that define WHAT to report are scanned images: pypdf extracts zero characters from Reg 8 (minimum instrumentation, s.32(1)), Reg 9 (forwarding analysis of readings to the SDSO, s.32(2)) and Reg 16 (hydro-met station data, s.33(1)). Someone must read or OCR them before AquaSync can claim a compliant cadence or form. Do not quote a reporting interval until that is done. The API itself is undocumented and can change without notice.

Regulatory. Dam Safety Act 2021 ss.32(1), 32(2), 33(1), 34(1), 34(2) - and note Reg 11 (s.34(2), 'location and manner of collection, compliance, process and storage of data') is a data-storage mandate directly relevant to where AquaSync's telemetry layer lives.

Idukki dam instrumentation and any KSEB OT estate

Approach. Do NOT attempt a field-bus client. The only realistic routes are (a) a negotiated export or read-only database view from Encardio's Drishti system, or (b) a read-only DMZ historian read over OPC-UA using opcua-asyncio (LGPL-3.0, 1,468 stars, actively pushed). Meanwhile build a SIMULATED IEC 60870-5-104 testbed with Fraunhofer's c104 (GPL-3.0) so the protocol layer is exercised without touching plant.

Protocol or standard. OPC-UA for a historian read; IEC 60870-5-104 (with IEC 62351 security extensions) is the protocol CEA names for RTU/PLC conformance; IEC 61850-7-410 defines hydropower logical nodes if a plant vocabulary is ever needed (libiec61850, GPL-3.0, though the hydro node model lives in the paid IEC standard).
Effort. High - quarters, and mostly negotiation.
Blockers. The incumbent stack is closed and vendor-proprietary: Encardio's RESHMI contract under DRIP, geotechnical/geodetic/hydro-met/seismic sensors, data loggers with GSM/GPRS to a central server, Drishti database management. No open protocol socket is evidenced anywhere. Earlier planning material asserted 73 telemetered stations and 50 SCADA substations on ICCP/TASE.2 to SLDC Kalamassery - that is UNSOURCED and could not be confirmed on any KSEB page; treat it as inference, never as fact. Any commercial-licence copyleft (GPL-3.0 on c104/libiec61850) is a real obligation for a KSEB-bound deliverable.

Regulatory. CEA Guidelines on Cyber Security in Power Sector 2021, cardinal principle: 'There is hard isolation of their OT Systems from any internet facing IT system', at most one internet-facing IT system per site under CISO control with data movement only via a whitelisted device. This is the quotable basis for a one-way/read-only architecture. CERT-In directions of 28-04-2022 add 6-hour incident reporting, 180-day rolling logs held within Indian jurisdiction, and NTP sync to NIC or NPL.

Publishing AquaSync's own model output to partner agencies

Approach. Serve AquaSync's time series through pygeoapi (MIT, 623 stars, PyPI 0.24.0) as an OGC API - EDR endpoint. EDR's position/radius/area/cube/trajectory queries are shaped precisely for 'give me the series at this point over this window' against model output, which is what a decision-support twin publishes. NDSA's own open JSON endpoints are the local precedent that agencies here do publish machine-readable data.

Protocol or standard. OGC API - Environmental Data Retrieval (OGC 19-086r6). OGC SensorThings (18-088, via FROST-Server, LGPL-3.0 + Java/PostGIS) only if a partner explicitly demands it. WaterML 2 Part 3 / HY_Features (14-111r6) if a catchment/reach vocabulary is needed.
Effort. Medium - 1-2 weeks for a read-only EDR facade.
Blockers. India has adopted none of these standards - this is a unilateral design choice, not a compliance requirement, so it buys interoperability rather than approval. pygeoapi requires Python >=3.12, a real constraint if AquaSync is pinned lower. 'OGC HydroDART' from the original planning material does not exist; do not let it creep back in.

Regulatory. Dam Safety Act 2021 s.35(1)(e) requires anticipated inflow/outflow information to be made available to district authorities AND 'in public domain' - a published EDR endpoint is a direct, defensible way to satisfy that.

CWRDM / UW-SASWE (RAT-Kerala) - the tractable research partner

Approach. Consume the published per-dam CSVs immediately (officially linked from cwrdm.kerala.gov.in, so sanctioned rather than scraped) for surface area, storage change, evaporation and an independent area-elevation curve. Separately, approach CWRDM as the single most tractable point of contact for in-situ KSEB 2018 inflow: three independent teams state they obtained it, and Suresh et al. say verbatim 'in-situ inflow data obtained from the KSEB for Idamalayaar reservoir' with Table 5 captioned 'Observed inflow from KSEB for the period of May - October 2018'.

Protocol or standard. Plain HTTP CSV at a fixed path pattern: /RAT_Kerala_CWRDM/assets/RAT_results/{product}/{dam}.csv. No auth.
Effort. Low to consume (hours). Medium to partner (weeks to months).
Blockers. Plain HTTP on a bare IP with no TLS and no DNS name - it can move without notice - and the page carries a live 'Data is still being populated' banner. RAT inflow is MODELLED VIC output, not observation: published Idukki skill is only r=0.63 (RMSE 217.94 cumec) and its 2018 Idukki peak of 1104.56 cumec on 18 Aug is 2.3x low and 3 days late against CWC's observed 2532 cumec at 22:00 on 15 Aug. The source paper itself says inflow peaks 'are found to be underpredicted... requiring bias correction'. Note the dam slug is idamalayaar with a double-a.

Regulatory. RAT source is GPL-3.0, which has consequences if any of it is reused inside a KSEB-bound AquaSync. Consuming the CSVs carries no such obligation.

5 · Historical Kerala / India Reservoir And Flood Data, Especially Pre 2020

Confirmed sources (30)

Source	Kind	What it provides	Relevance	HTTP
National Water Data Portal (NWDP) CKAN API metadata_m · Other (Open) https://nwdp.nwic.gov.in/api/3/action/package_list	open API (CKAN 2.11.3)	package_list returns exactly 555 packages; status_show reports ckan_version 2.11.3, site_title 'National Water Data Portal'. package_show?id= returns direct, unauthenticated resource…	Highest. This is the door to genuine pre-2020 Periyar gauge data and is absent from the…	200
CWC Kerala daily river discharge 2001-2025 (NWDP resource) https://nwdp.nwic.gov.in/dataset/08fa3fd0-7861-471d-a295-…	CSV bulk download (63 MB pair with the 1950-2000 file)	I downloaded it: 39,425,558 bytes, 217,538 rows, 40 Kerala CWC stations, 30,833 Periyar rows - all exact. Three Periyar gauges confirmed with coordinates: NEELEESWARAM (10.18361, 76.49556) n=8478,… caveat: The Aug-2018 gap is real but LARGER than reported: Neeleeswaram has 18 of 31 days, missing days 5, 12, 16-22 AND 24-27 (13 days, not the 7…	Highest - closes part of gap #1. Neeleeswaram is the exact CWC G&D; site AquaSync routes…	200
CWC Kerala daily river discharge 1950-2000 (NWDP resource) https://nwdp.nwic.gov.in/dataset/08fa3fd0-7861-471d-a295-…	CSV bulk download	Downloaded: 23,703,774 bytes, 134,069 rows, 21 stations, 19,588 Periyar rows - all exact. NEELEESWARAM n=10,884 starting 1971; ARANGALI n=8,276 from 1978; VANDIPERIYAR n=428 (year 2000 only)…. caveat: CORRECTION: the sweep claimed Neeleeswaram goes to '730/yr from 1978 as two readings per day'. It does not. Per-year counts are 291 (1971)…	High - this is the deep historical record, half a century of daily Periyar discharge at…	200
Kerala SW manual daily rainfall (NWDP resource, mislabelled 1991-2020) https://nwdp.nwic.gov.in/dataset/c379c7dd-755c-4a0d-9de1-…	CSV bulk download	Downloaded: 24,200,224 bytes, 151,429 rows - exact. Real year span is 2000-2020, NOT 1991 as the filename says. 2018 holds 8,435 station-days; IDUKKI district 7,462 rows; ERNAKULAM 366 - all exact.… caveat: TWO corrections. (1) The sweep calls BOYCE ESTATE and Vannapuram 'long-record catchment stations'. They are Idukki DISTRICT stations, not…	Moderate - useful gauge rainfall for SCS-CN antecedent moisture, but see the caveat…	200
NWDP Kerala Surface Water Dept river discharge - VERIFIED NEGATIVE, label is false https://nwdp.nwic.gov.in/dataset/river-discharge-manual-d…	negative result / trap to avoid	Advertised in CKAN as 'River Discharge Kerala_SW Kerala (2001 - 2025) Manual Daily'. I downloaded the resource: 765,060 bytes, 4,904 rows, every single row from calendar year 2011, 23 stations, ZERO…	High as a warning. This is exactly the kind of confidently-mislabelled dataset that…	200

Source	Kind	What it provides	Relevance	HTTP
RAT 3.0 Kerala live dashboard (CWRDM + UW SASWE + NASA) http://210.212.226.241:8080/RAT_Kerala_CWRDM/	static-file dashboard, no auth	Page confirms the six product families exactly as claimed: Inflow, Outflow, Surface Area, Storage Change, Area-Elevation Curve, Evaporation. Footer reads 'Maintained by CWRDM in collaboration with… caveat: Plain HTTP on a bare IP (no TLS, no DNS name), so it can move without notice - though cwrdm.kerala.gov.in links to this exact URL, so it…	High. Mullaperiyar in particular matters - Suresh et al. show it regulates what actually…	200
RAT Kerala - Idukki and Idamalayar daily inflow CSVs http://210.212.226.241:8080/RAT_Kerala_CWRDM/assets/RAT_r…	CSV time series	Downloaded: 100,264 bytes, 3,410 rows, columns date,streamflow, 2015-10-24 to 2025-02-22. I checked continuity programmatically - ZERO non-one-day steps, genuinely unbroken. August 2018 complete at… caveat: The sweep's honesty here is warranted and I confirmed the specific numbers: RAT's 2018 Idukki peak is 1104.56 m3/s on 2018-08-18, against…	High for continuity, but read the caveat before using it as truth for 2018.	200
RAT Kerala - Idukki and Idamalayar area-elevation curves http://210.212.226.241:8080/RAT_Kerala_CWRDM/assets/RAT_r…	CSV lookup table	Idukki: 626 rows at 1 m steps, 551-1176 m, columns elevation,area. Verified values 50.015 km2 @730 m, 51.098 @731, 52.125 @732, 53.201 @733. Idamalayar (aec/idamalayaar.csv): 588 rows, 62-649 m,… caveat: Satellite/DEM-derived, not a KSEB survey curve. The Suresh et al. paper validates AEC shape against KSEB observed curves only for…	High and directly actionable - an independent area-elevation curve to cross-check…	200
RAT Kerala - Idukki TMS-OS surface area and storage change http://210.212.226.241:8080/RAT_Kerala_CWRDM/assets/RAT_r…	CSV time series	657 rows, 2017-07-29 to 2025-02-21, columns date,'area (km2)', with 85 observations in 2018. Paired /dels/idukki.csv is also 657 rows with columns date,'dS (m3)'. Both confirmed by download.	Moderate - an independent, observation-based (not modelled) handle on 2018 storage…	200
CWC Reservoir Level & Storage Bulletin archive https://cwc.gov.in/en/reservoir-level-storage-bulletin	PDF archive, paginated	I scraped pages 0-25 and extracted 528 distinct PDF links. Per-year counts: 2015:36, 2016:51, 2017:51, 2018:48, 2019:50, 2020:52, 2021:50, 2022:50, 2023:43, 2024:46, 2025:15. Weekly Thursday… caveat: The archive effectively BEGINS in 2015. Pre-2015 is one or two stray PDFs per year (2001:1, 2002:1, ... 2012:2), not a weekly series. So…	Highest for gap #1 - this is the only real, official, numeric 2018 Idukki level+storage…	200
CWC weekly bulletin, 02 August 2018 (worked example) https://cwc.gov.in/sites/default/files/02.08.2018cwcbull_…	PDF, 7 pages	539,052 bytes, exact. I extracted the rows verbatim: '*82 IDUKKI 732.43 730.25 1.460 1.338 01-08-2018 92 23 41 - 780' and, as a bonus the sweep did not mention, '*81 IDAMALAYAR 169.00 167.03 1.018… caveat: BONUS FINDING: the Idamalayar row independently confirms AquaSync's second reservoir too - FRL 169.00 m, live capacity 1.018 BCM,…	Highest. A free, official cross-check on AquaSync's hard-coded constants, and the only…	200

Source	Kind	What it provides	Relevance	HTTP
CWC 'Kerala Floods of August 2018' study report (hosted by KSDMA) https://sdma.kerala.gov.in/wp-content/uploads/2020/08/CWC…	PDF, 48 pages	6,003,314 bytes and 48 pages, both exact. I extracted and confirmed every quantitative claim: Idukki peak inflow 2532 cumec at 22:00 on 15-08-2018; corresponding release 1614 cumec (1500 spill + 114… caveat: Corroborates the gauge-failure story precisely: CWC states the 8800 cumec peak occurred on 16 August at 15:00, and the NWDP CWC gauge file…	Highest. This is the primary source for the flagship 2018 case study, and it is…	200
CWC 2018 report Fig.4 - the 6-hourly Idukki hydrograph, recoverable as vector geometry https://sdma.kerala.gov.in/wp-content/uploads/2020/08/CWC…	vector chart embedded in PDF (page 13)	I located Fig.4 'Inflow, Outflow and Water Level: Idukki Reservoir' on page index 12 (page 13) and inspected it with PyMuPDF: 2,571 vector path items on that page. The axis tick labels are live PDF… caveat: The page ALSO contains 20 raster images, so it is a mixed page - a path extractor must separate the plotted series from raster decoration.…	Highest for the 2018 gap. This is the closest thing to a sub-daily 2018 Idukki…	200
Sudheer et al. 2019, 'Role of dams on the floods of August 2018 in Periyar River Basin, Kerala', Current Science 116(5):780-794 https://www.currentscience.ac.in/Volumes/116/05/0780.pdf	peer-reviewed paper, free full PDF	4,413,605 bytes, 15 pages, both exact. Authors confirmed (K. P. Sudheer, S. Murty Bhallamudi, Balaji Narasimhan, Jobin Thomas, V. M. Bindhu, Vamsikrishna Vema, Cicily Kurian - IIT Madras / Purdue).… caveat: Table 1 extracts with SHIFTED COLUMNS - dam names do not line up with their type/year/length/height values in extracted text (e.g.…	Highest. A published, independent HEC-HMS reconstruction of the exact basin AquaSync…	200
Suresh et al. 2024 (Remote Sensing of Environment) - free accepted manuscript on NASA NTRS https://ntrs.nasa.gov/api/citations/20240007715/downloads…	accepted manuscript PDF, 56 pages, open	3,714,034 bytes, 56 pages, both exact. Free copy of the paywalled RSE paper (Crossref confirms DOI 10.1016/j.rse.2024.114149, RSE vol 307 art 114149, 2024-06, 21 citations). Verified: Table 1 lists… caveat: KSEB provenance confirmed verbatim at two places: 'in-situ inflow data obtained from the KSEB for Idamalayaar reservoir' and the Table 5…	Highest. It is both the published skill statement for the RAT CSVs AquaSync would…	200
Minocha et al. 2024, 'Reservoir Assessment Tool version 3.0', Geoscientific Model Development 17:3137-3156 Creative Commons (Copernicus open access) https://gmd.copernicus.org/articles/17/3137/2024/	open-access journal article	Crossref-verified: published 2024-04-23, GMD vol 17 pages 3137-3156, 6 citations. Abstract confirmed on-page: RAT is 'a data-driven software platform that integrates satellite remote sensing with…	Moderate - the methods reference you cite when using any RAT-Kerala CSV, and the place…	200
UW-SASWE/RAT (Reservoir Assessment Tool source) 26★ · pushed_at · GPL-3.0 https://github.com/UW-SASWE/RAT	GitHub repository (Python)	GitHub API confirms: full_name UW-SASWE/RAT, 26 stars, 6 forks, created 2022-05-12, pushed 2026-01-13, not archived, GPL-3.0, 27,764 KB, description 'Reservoir Assessment Tool'. This is the…	Moderate - only worth pulling if AquaSync wants to re-run RAT with bias correction…	200

Source	Kind	What it provides	Relevance	HTTP
IMD 0.25-degree daily gridded rainfall - direct POST endpoint https://www.imdpune.gov.in/cmpg/Griddata/RF25.php	unauthenticated POST download endpoint	I ran the POST live with RF25=2018: HTTP 200, 25,431,832 bytes, exact. Magic bytes 'CDF\x01' (NetCDF classic), numrecs 365. Parsed with scipy: LONGITUDE(135), LATITUDE(129), TIME(365, 'days since… caveat: 0.25 deg is ~28 km - a single cell spans much of the Idukki catchment, so this is basin-scale forcing, not sub-catchment. 2018 is a…	Highest for forcing. One unauthenticated line gives basin rainfall for any year…	200
IMD gridded rainfall NetCDF landing page https://www.imdpune.gov.in/cmpg/Griddata/Rainfall_25_NetC…	HTML form / documentation page	Confirmed: form action is 'RF25.php' with select name 'RF25', and the dropdown carries 126 options - 'Select' plus every year from 2025 down to 1901. Page text states 'Long Period (1901-2024) Daily… caveat: Minor: the dropdown actually offers 2025 as well, one year beyond the documented 1901-2024 span.	Low on its own but it documents the POST contract and the grid layout, so it is the…	200
imdlib (Python client for IMD gridded data) latest rel · MIT https://pypi.org/project/imdlib/	PyPI package	PyPI JSON confirms version 0.1.21, MIT licence, requires_python >=3.0, summary 'A tool for handling and downloading IMD gridded data', 28 releases, latest upload 2025-12-04. Home page…	Moderate convenience - wraps the IMD download and parses both the NetCDF and legacy…	200
GUARDIAN: ExtendinG SUb-DAily River Discharge data over INdia (Scientific Data, 2024) https://doi.org/10.1038/s41597-024-03923-8	peer-reviewed data descriptor	Crossref-verified: 'ExtendinG SUb-DAily River Discharge data over INdia (GUARDIAN)', Scientific Data vol 11, published 2024-10-18, 5 citations. Describes hourly monsoon / sub-daily non-monsoon… caveat: The sweep says '210 CWC stations'; the rating-curve file I downloaded contains exactly 210 data rows, which corroborates it.	High for gap #2 (Muskingum K/x calibration) and only that. It starts in 2020, so it does…	200
GUARDIAN data on figshare (data.rar + rating curves) https://doi.org/10.6084/m9.figshare.27004282	figshare data deposit	figshare API (article 27004282) confirms: title matches, published 2024-10-19, DOI 10.6084/m9.figshare.27004282.v1, two files - data.rar 577,023,190 bytes and RC.xlsx 53,652 bytes. I downloaded… caveat: COORDINATE CONFLICT I found that the sweep missed. GUARDIAN places VANDIPERIYAR at (9.1525, 77.1525), which is in Tamil Nadu, roughly 47…	High for gap #2. Hourly Neeleeswaram discharge from 2020 overlaps the amith-vp KSEB…	200
GHI: Geospatial dataset for hydrologic analyses in India (ESSD 15:4389, 2023) https://essd.copernicus.org/articles/15/4389/2023/	open-access data descriptor + dataset	Crossref-verified 2023-10-05, ESSD vol 15 pages 4389-4415, 17 citations. Abstract confirmed on-page: of CWC's 645 Peninsular-India stations, Group 1 = 213 (reliable metadata + adequate daily… caveat: UPGRADE, not a correction: the sweep says GHI merely 'mentions the Periyar River'. In fact GHI uses NEELEESWARAM ON THE PERIYAR as its…	High - and more directly relevant than the sweep realised (see caveat). Gives…	200

Source	Kind	What it provides	Relevance	HTTP
CAMELS-IND (ESSD 17:461, 2025) https://essd.copernicus.org/articles/17/461/2025/	open-access data descriptor + dataset	Crossref-verified 2025-02-05, ESSD vol 17 pages 461-491, 23 citations. Abstract confirmed on-page verbatim: 472 catchments in Peninsular India of which 228 have observed streamflow for over 30% of… caveat: The paper's own TITLE says '228 catchments' while the abstract says 472 with 228 gauged - the sweep's framing is correct but be aware the…	Moderate - a ready-made benchmark and a source of catchment attributes (including dam…	200
Mishra et al., 'The Kerala flood of 2018: combined impact of extreme rainfall and reservoir storage' (HESS Discussions preprint) https://hess.copernicus.org/preprints/hess-2018-480/	preprint - NOT peer-reviewed, see caveat	Preprint page and full PDF (1,301,376 bytes) both retrieved. Content claims confirmed verbatim: 'Six out of seven major reservoirs were at more than 90% of their full capacity on August 8, 2018';… caveat: STATUS PROBLEM the sweep did not report: the page states 'The manuscript was not accepted for further review after discussion.' This is a…	Moderate - its real value is the pointer in the caveat, not its conclusions.	200
India-WRIS API contract, recovered from WRIS_India_data_extractor source 0★ · pushed_at · none declared https://github.com/DVTanna369/WRIS_India_data_extractor	GitHub repository (Flask app) - value is the documented contract	GitHub API confirms the repo exists: created 2025-08-27, pushed 2025-09-01, not archived, 58 KB. I read main_app/routes.py, api_client.py and helpers.py and confirmed EVERY endpoint claimed, at the… caveat: TWO problems the sweep did not flag. (1) indiawris.gov.in (164.100.85.36) is UNREACHABLE from here - DNS resolves but TCP connect to port…	Potentially high (India-WRIS is where Mishra et al. got 2007-2018 Idukki storage, and…	200
amith-vp/Kerala-Dam-Water-Levels (already in AquaSync - refreshed metrics) 54★ · pushed_at · none declared https://github.com/amith-vp/Kerala-Dam-Water-Levels	GitHub repository (scraped JSON feed)	GitHub API confirms exactly as claimed: 54 stars, 8 forks, pushed_at 2026-08-26T08:05:16Z (today), size 112,128 KB, created 2024-07-02, not archived. Description: 'Repository of Live and Historical… caveat: Nothing here contradicts the established project facts: this feed still covers 2020-08-13 to 2026-08-26 only and contains no 2018 data,…	Already integrated. Its value in this sweep is as the 2020-onward partner for GUARDIAN…	200
GloLakes: water storage dynamics for 27,000 lakes (ESSD 16:201, 2024) https://essd.copernicus.org/articles/16/201/2024/	open-access data descriptor + dataset	Crossref-verified: 'GloLakes: water storage dynamics for 27 000 lakes globally from 1984 to present derived from satellite altimetry and optical imaging', ESSD vol 16 pages 201-218, published… caveat: UNVERIFIED: I did not confirm that Idukki or Idamalayar are among the 27,000 lakes, nor that either has usable altimetry (Idukki is a…	Moderate and speculative for AquaSync - global rather than Kerala-specific, but it is…	200
KSDMA dam water level page and 2018 floods section https://sdma.kerala.gov.in/dam-water-level/	government portal	Kerala State Disaster Management Authority daily water-level page for major KSEB and Irrigation dams. The /floods_2018/ section also returns 200, and KSDMA is the host that actually serves the CWC…	Moderate - and it is the natural institutional counterpart to CWRDM for gap #5…	200

Source	Kind	What it provides	Relevance	HTTP
CWRDM - Centre for Water Resources Development and Management, Kerala https://cwrdm.kerala.gov.in/reservoir-assessment-tool-ker…	state R&D; institution page	Page text confirms verbatim: 'RAT 3.0 Kerala was co-developed by CWRDM and the SASWE Research Group at the University of Washington, USA with support from NASA Applied Science - Earth Action Water… caveat: This link is what elevates the bare-IP RAT dashboard from 'found on the internet' to 'officially published by a Kerala state institute'….	Highest for gap #5 and for the 2018 gap. Three independent teams state they obtained…	200

Verification notes

- METHOD: all 30 URLs curl-checked; 28 returned 200, two returned 202 (doi.org->figshare, which resolves fine via the figshare API, and dataverse.tdl.org, which does not). I went past status codes: I downloaded and parsed the three NWDP CSVs (87 MB), the four PDFs, the eight RAT CSVs, the IMD 2018 NetCDF and the GUARDIAN rating-curve workbook, and re-computed every headline number. The sweep is…
- CORRECTION 1 - the flood peak WAS measured, just not at Neeleeswaram. The sweep's headline caveat says 'the peak was never measured'. That is true for NEELEESWARAM (18 of 31 August days; and the gap is worse than reported - 13 days missing: 5, 12, 16-22, 24-27, not the 7 claimed) and equally true for ARANGALI (missing 16-27). But VANDIPERIYAR, the upstream Periyar headwater gauge, recorded ALL…
- CORRECTION 2 - the 1950-2000 discharge file is one reading per day throughout. The sweep claims Neeleeswaram goes to '730/yr from 1978 as two readings per day'. Per-year counts are 291 (1971) then 365 or 366 for every single year 1972-2000. No year has 730. The twice-daily pattern (08:00 and 08:30, or 09:00/09:15) only appears in the 2001-2025 file.
- CORRECTION 3 - district is not catchment. BOYCE ESTATE and Vannapuram are described as 'long-record catchment stations' inside the Idukki catchment. Both are in Idukki DISTRICT, not the Idukki reservoir catchment: BOYCE ESTATE is in Peerumade tehsil at 9.565N on the southern/western slope, and Vannapuram's stored longitude (76.463E) sits far west of Thodupuzha town (~76.72E), so its coordinate…
- CORRECTION 4 - a station coordinate conflict between two of the sweep's own sources. GUARDIAN places VANDIPERIYAR at (9.1525, 77.1525), across the border in Tamil Nadu; NWDP/CWC places it at (9.57333, 77.09056), the correct site of Vandiperiyar town in Kerala - a ~47 km discrepancy. NEELEESWARAM (10.18361, 76.49556) and ARANGALI (10.28611, 76.32222) agree EXACTLY between the two datasets, which…
- CORRECTION 5 - the HESS preprint was REJECTED. Its page states 'The manuscript was not accepted for further review after discussion.' The sweep presents it as an 'open preprint' and repeats its >500-year return period finding without qualification. Worse, that figure directly contradicts the peer-reviewed Current Science paper in the same list, which says the Periyar basin received a 145-YEAR…
- CORRECTION 6 - India-WRIS is unreachable, so the 'recovered API contract' is documentation, not access. The contract itself is transcribed faithfully (I verified all seven endpoints line by line in the repo source). But indiawris.gov.in (164.100.85.36) times out on TCP connect to :443 after ~21 s on every endpoint tried. The sweep reported '200' for the GitHub repo, which is correct but easy to…
- SCOPE CORRECTION - the CWC bulletin archive starts in 2015, not earlier. Scraping pages 0-25 yielded 528 PDFs: 36 for 2015, ~50/yr for 2016-2024, and only one or two stray files per year for 2001-2012. So this gives weekly Idukki level+storage for 2015-2025 and nothing usable before that. The '45 of 52 for 2018' claim is exactly right - I enumerated the Thursdays and confirmed the 7 absentees…
- TWO BONUSES the sweep under-sold. (1) The bulletins validate BOTH reservoirs, not just Idukki: the same PDFs carry '*81 IDAMALAYAR 169.00 167.03 1.018 0.961 01-08-2018 94 30 45 33 75', independently

confirming AquaSync's Idamalayar FRL of 169.00 m and 75 MW alongside Idukki's 732.43 m / 1.460 BCM / 780 MW - four project constants cross-checked from one official PDF. (2) GHI does not merely…

- PRACTICAL WARNING for whoever ingests the papers: BOTH the Current Science and NTRS PDFs extract with SHIFTED TABLE COLUMNS. In Current Science Table 1 the dam names do not line up with their type/year/length/height values (one row loses its Type, another its name), so the '14 dams' count is approximate. In the NTRS manuscript Table 1 the Sl.No/reservoir-name column is offset by one row against…

Rejected (1)

Recorded so the same ground is not covered twice.

Claimed source	Why it was rejected
Global Reservoir Surface Area Dataset (GRSAD)	Cannot be obtained, and would not help if it could. dataverse.tdl.org returns HTTP 202 with a ZERO-BYTE body to every non-browser client - the landing page, the DOI redirect target, and its own native API…

6 · Open Source Reservoir Operation, Optimisation And Forecast Informed Operations (Firo)

Confirmed sources (24)

Source	Kind	What it provides	Relevance	HTTP
wltaylor/Robust-FIRO 1★ · pushed 202 · MIT https://github.com/wltaylor/Robust-FIRO	GitHub repo (research code for a published paper)	MPCProblem.py verified at EXACTLY 194 lines. Objective confirmed line-by-line: Sf = S0 + cp.cumsum(Qf - u, axis=1) over an (NE x NL) ensemble-by-lead array; cost = (1-w)*cp.sum(cp.pos(Sf-TOCS)) +… caveat: Two frictions the source list omitted. (1) Line 4 is a bare, unguarded `import mosek` and run() defaults to solver='MOSEK', a COMMERCIAL…	Highest-value item in the sweep and it survives adversarial checking intact. The…	200
wltaylor/SSJRB-FIRO-Publishing 1★ · pushed 202 · MIT https://github.com/wltaylor/SSJRB-FIRO-Publishing	GitHub repo (research code for a published manuscript)	Multi-reservoir FIRO across 14 Sacramento-San Joaquin reservoirs. All claimed files verified present: train_firo.py (3,492 B), perfect_forecast.py (1,883 B), simulate_baseline_tocs.py (2,679 B),… caveat: CORRECTION: perfect_forecast.py is NOT 'an explicit perfect-foresight baseline isolated so the gap can be measured'. I read all 1,883…	The risk-threshold policy parameterisation is a cheaper alternative to full MPC and the…	200
zpb4/syn-forecast_stochastic (Brodeur) 1★ · pushed 202 · MIT https://github.com/zpb4/syn-forecast_stochastic	GitHub repo (research code, manuscript in review)	Synthetic hydrology generation from a Stochastic Weather Generator + SAC-SMA, synthetic ensemble-forecast fitting, deliberate forecast-SKILL MODIFICATION (improvement/degradation), and a FIRO model… caveat: NOT pure Python, which the source list implied. Declared dependencies are a MIXED R + Python stack: R packages 'future', 'DEoptim',…	The skill-degradation machinery is the honest way to answer 'how good does an Idukki…	200
rtc-tools/rtc-tools 24★ · pushed 202 · LGPL-3.0 https://github.com/rtc-tools/rtc-tools	GitHub repo (Deltares framework)	Verified by shallow clone: src/ contains EXACTLY 16,410 lines of Python across 46 files. ControlTreeMixin confirmed at src/rtctools/optimization/control_tree_mixin.py, docstring 'Adds a stochastic… caveat: The Modelica lock-in verdict is CONFIRMED by exhaustive per-example check. All 13 optimisation examples import ModelicaMixin; the only two…	Two designs worth stealing without adopting the framework: the goal-programming priority…	200
rtc-tools on PyPI 2.8.0 rele · LGPL-3.0 https://pypi.org/project/rtc-tools/	PyPI package page	Verified from the PyPI JSON API: version 2.8.0, uploaded 2026-07-08T15:14:41, requires-python >=3.10, 88 releases, LGPL-3.0. requires_dist verified exactly as claimed: casadi!=3.6.6,<3.8,>=3.6.3;… caveat: The claimed CasADi bundled-solver enumeration (IPOPT, Bonmin, HiGHS, CBC, Clp, qpOASES present; Knitro and Gurobi absent) could NOT be…	Confirms the low dependency burden claim at the metadata level - 7 hard runtime deps, no…	200

Source	Kind	What it provides	Relevance	HTTP
Lake Mendocino FIRO Final Viability Assessment (Dec 2020) published https://cw3e.ucsd.edu/FIRO_docs/LakeMendocino_FIRO_FVA.pdf	PDF report (141 pages, 11.3MB, application/pdf, %PDF-1.7)	Verified by extracting all 297,890 characters of text. PDF /Title metadata: 'Lake Mendocino Forecast Informed Reservoir Operations: Final Viability Assessment'; /Author 'FIRO Steering Committee';… caveat: One timing nuance: 'FIRO is now authorised in the USACE Water Control Manual' overstates what THIS document says. The Dec 2020 FVA…	The evidence base for arguing FIRO to KSEB/KSDMA: a regulator-grade document showing a…	200
Prado Dam FIRO Final Viability Assessment (2023) published https://cw3e.ucsd.edu/FIRO_docs/Prado/FIRO_Prado_FVA.pdf	PDF report (136 pages, 10.0MB, application/pdf, %PDF-1.7)	Verified by extracting all 300,856 characters. Created 2023-11-28. Prado Dam Steering Committee co-chaired by Orange County Water District (Greg Woodside 2017-2023, then Adam Hutchinson… caveat: FABRICATED FIGURE: the string '23,000' appears ZERO times in the entire 300,856-character document. The claim 'up to 23,000 acre-feet of…	Second completed FVA and the closest FIRO analogue to a conjunctive-use objective.…	200
CW3E FIRO programme portal live page, https://cw3e.ucsd.edu/firo/	Institutional programme index page	Fetched and text-extracted (80,605 bytes). Confirms exactly six FIRO projects in its navigation and body: Howard A. Hanson Dam, Prado Dam, Russian River Watershed, Seven Oaks Dam, Willamette Valley,… caveat: Page names the programme 'Willamette Valley' in the project list; the source list shortened it to 'Willamette'. No substantive difference.	Confirms the negative that matters for planning: FIRO is a real, deployed, multi-site…	200
pywr/pywr 191★ · pushed 202 · GPL-3.0 https://github.com/pywr/pywr	GitHub repo + PyPI package	Verified via GitHub code search that all claimed components exist: HydropowerRecorder (pywr/recorders/_hydropower.pyx), HydropowerTargetParameter (pywr/parameters/_hydropower.pyx),… caveat: GPL-3.0 is viral, which matters if anything is destined for KSEB. Small addition: the [opt] extra pulls pygmo as well as platypus-opt -…	Adopt the CONCEPTS not the framework, as the prior agent concluded - the ControlCurve…	200
Pywr-DRB/Pywr-DRB 12★ · pushed 202 · MIT https://github.com/Pywr-DRB/Pywr-DRB	GitHub repo (applied basin model)	'Water management model for the Delaware River Basin, based on the pywr simulation framework' - a real, maintained basin-scale reservoir-operations application, not a demo. Also independently listed… caveat: Small project (12 stars). The specific claims about NYC reservoirs, transbasin diversions and the 1954 Decree Montague target were not…	The best worked example of what a serious multi-reservoir regulatory-rule model looks…	200

Source	Kind	What it provides	Relevance	HTTP
iRONStoolbox/iRONS 11★ · pushed 202 · MIT https://github.com/iRONStoolbox/iRONStoolbox	GitHub repo (teaching + research toolbox)	Every claimed module verified present under iRONS/Software/ with matching sizes: res_sys_sim.py (13,443 B), operating_policy.py (12,586 B), HBV_sim.py (7,133 B), bias_correction.py (6,614 B),… caveat: Default branch is 'master', not 'main' - raw.githubusercontent URLs must use master. Copernicus CDS access requires a free account and API…	op_piecewiselin_1res is the single most directly reusable artefact in the whole sweep…	200
anyoptimization/pymoo 2946★ · pushed 202 · Apache-2.0 https://github.com/anyoptimization/pymoo	GitHub repo + PyPI package	Multi-objective optimisation framework. PyPI 0.6.2, requires-python >=3.10. Dependencies verified exactly as claimed: numpy>=1.19.3, scipy>=1.1, moocore>=0.1.7, autograd>=1.4, cma>=3.4.0,… caveat: The head-to-head benchmark claim (grid 4,800 evals in 2.02 s vs NSGA-II 2.31 s on identical budget) could NOT be reproduced - pymoo is not…	The correct MOEA choice for AquaSync precisely because it is Apache-2.0 - it is the…	200
Project-Platypus/Platypus 654★ · pushed 202 · GPL-3.0 https://github.com/Project-Platypus/Platypus	GitHub repo	'A Free and Open Source Python Library for Multiobjective Optimization'. Confirmed as the optional dependency behind Pywr's [opt] extra (pyproject.toml: opt = ["platypus-opt", "pygmo"], and also in… caveat: GPL-3.0. The 'substrate PyBorg is built on' claim was not independently verified and inherits Borg's access restrictions in any case -…	Relevant mainly as the substrate other tools reach for, and as the concrete link between…	200
DEAP/deap 6436★ · pushed 202 · LGPL-3.0 https://github.com/DEAP/deap	GitHub repo + PyPI package	Distributed Evolutionary Algorithms in Python. PyPI 1.4.4. Runtime dependencies verified to be exactly two: numpy and moocore. caveat: General-purpose EC toolkit with no reservoir or water-resources content of its own.	Low direct relevance to AquaSync (pymoo covers the same ground with a higher-level API),…	200
MOEAFramework/MOEAFramework 361★ · pushed 202 · LGPL-3.0-or-later https://github.com/MOEAFramework/MOEAFramework	GitHub repo (Java framework)	Java multi-objective optimisation framework. Cornell Reed Group page independently confirms it supports GDE3, MOEA/D, NSGA-II, epsilon-MOEA, epsilon-NSGA-II and random search across 'over 80 test… caveat: LICENCE CORRECTION: the GitHub API reports NOASSERTION because the licence lives in a file named COPYING. I read it - it is explicitly…	Reference value for algorithm diagnostics methodology rather than something to adopt -…	200
odow/SDDP.jl 355★ · pushed 202 · MPL-2.0 https://github.com/odow/SDDP.jl	GitHub repo (Julia/JuMP package)	Multistage stochastic dual dynamic programming as a JuMP extension. Backing paper independently confirmed via the Crossref API: DOI 10.1287/ijoc.2020.0987 = 'SDDP.jl: A Julia Package for Stochastic… caveat: LICENCE CLARIFICATION: GitHub API reports NOASSERTION; the LICENSE.md file states plainly 'SDDP.jl is licensed under the Mozilla Public…	The right tool for the OTHER half of the foresight problem - the seasonal question 'how…	200

Source	Kind	What it provides	Relevance	HTTP
do-mpc/do-mpc 1443★ · pushed 202 · LGPL-3.0 https://github.com/do-mpc/do-mpc	GitHub repo	'Model predictive control python toolbox' built on CasADi/IPOPT, covering nonlinear MPC and moving-horizon estimation. caveat: Nearly 10 months without a push at time of verification - not dead, but slowing. Much heavier than the ~40-line cvxpy MPC that actually…	A fallback if the cvxpy formulation from Robust-FIRO proves too restrictive and AquaSync…	200
mxgiuliani00/M3O-Multi-Objective-Optimal-Operations 44★ · pushed 201 · GPL-2.0 https://github.com/mxgiuliani00/M3O-Multi-Objective-Optim…	GitHub repo (MATLAB toolbox)	Verified by directory listing - every claimed method has its own implementation directory, side by side: DDP (deterministic dynamic programming), SDP (stochastic DP), ISO (implicit stochastic… caveat: MATLAB and unmaintained since 2019 - read it, do not depend on it. GPL-2.0. The claim that EMODPS uses RBF policies was not verified…	Unmatched as a comparative map of the reservoir policy-design solution space - the…	200
Critical-Infrastructure-Systems-Lab/InfeRes 16★ · pushed 202 · MIT https://github.com/Critical-Infrastructure-Systems-Lab/In…	GitHub repo (Python package)	'Python package for inferring reservoir water extent, level and storage volume' from satellite imagery. caveat: Belongs to the data dimension, not this one - flagged here only because gap #1 is explicitly listed as a research target. Accuracy over…	Out of dimension but genuinely the most promising route to AquaSync's KNOWN GAP #1 - no…	200
Anurag14/Inflow-Prediction-Bhakra 17★ · pushed 202 · MIT https://github.com/Anurag14/Inflow-Prediction-Bhakra	GitHub repo (Indian reservoir, paper code)	Repo description confirms: 'Code for papers Optimal Reservoir Operations using Long Short-Term Memory Network and Bi-directional Storage Capacity and Elevation Level Calculator for Reservoir… caveat: Do not expect reusable code. Different basin (Satluj, not Periyar), different problem (inflow forecasting only, no operations…	Its real value is as EVIDENCE for a negative: it is the closest thing to Indian…	200
Reed Group software index (Cornell) live page, https://reed.cee.cornell.edu/software/	Curated institutional software index	Fetched and text-extracted (102,083 bytes). All eleven claimed entries confirmed present on the page: Borg MOEA, MOEA Framework, OpenMORDM, Hymod, HBV, Project Platypus, PySedSim, SALib, WaterPaths,…	The single best map of the water-resources MOEA ecosystem, and the fastest way to see…	200
Water Programming blog (Reed group) posts thro https://waterprogramming.wordpress.com/	Practitioner blog	Live and actively publishing - recent posts dated through 2026-08-19 include 'Introducing SynHydro: a Python package providing access to many synthetic streamflow generation methods', 'Using…	The practical how-to layer the papers omit. Two recent posts are unexpectedly on-target…	200
baobabsoluciones/flowing-basin 5★ · pushed 202 · MIT https://github.com/baobabsoluciones/flowing-basin	GitHub repo (paper code)	Intraday multi-reservoir hydropower economic optimisation - the authors' 'Hydropower Reservoirs Intraday Economic Optimization (HRIEO)' problem, maximising revenue. Three methods implemented side by… caveat: Default branch is 'master', not 'main'. Datasets, MILP/PSO solution files and RL model weights are all hosted on OneDrive links rather…	The closest published analogue to AquaSync's revenue-side problem (intraday,…	200

Source	Kind	What it provides	Relevance	HTTP
cylin928/HydroCNHS 22★ · pushed 202 · BSD-3-Clause https://github.com/cylin928/HydroCNHS	GitHub repo (Python package)	'A Python Package of Hydrological Model for Coupled Natural-Human Systems'. Verified by cloning: the four agent APIs are all genuinely present - Dam, RiverDiv, Conveying and InSitu. DEAP-based… caveat: CORRECTION: HydroCNHS does NOT offer HYMOD. I cloned the repo and grepped case-insensitively across every file - zero occurrences of…	Moderate. The agent-API pattern (a Dam agent embedded in a semi-distributed hydrological…	200

Verification notes

- FABRICATED NUMBER - Prado FVA. The claim 'up to 23,000 acre-feet of additional recharge in wet years' is NOT in the document cited. I downloaded the PDF and extracted all 300,856 characters; the string '23,000' occurs ZERO times. The document's actual key findings are 4,000-6,000 ac-ft/year additional groundwater recharge at buffer-pool elevations 508-512 ft, about 12,000 ac-ft/year at 520 ft,…
- MISDESCRIBED FILE - SSJRB perfect_forecast.py. Claimed to be 'an explicit perfect-foresight baseline - exactly AquaSync's current assumption, isolated so the gap can be measured'. It is not. I read the whole 1,883-byte file: it is a data-preparation script that builds a perfect-forecast netCDF by copying observed inflows forward into a (site, time, ensemble, lead) array, then plots and writes…
- FACTUAL ERROR - HydroCNHS hydrology options. Claimed 'GWLF, ABCD, HYMOD'. I cloned the repo and grepped case-insensitively across every file: zero occurrences of 'hymod'. Only GWLF and ABCD exist. The four agent APIs (Dam, RiverDiv, Conveying, InSitu) and the DEAP calibration harness are confirmed genuine.
- RESEARCH CODE, NOT LIBRARIES - three undisclosed frictions in the FIRO repos that the 'ready to port in 1-2 days' framing glosses over. (1) Robust-FIRO MPCProblem.py line 4 is a bare, unguarded `import mosek` and run() defaults to solver='MOSEK', a COMMERCIAL solver; CLARABEL is a real supported branch (line 84) and does work, but you must pass it explicitly AND still pip install mosek or delete…
- ALL SELF-REPORTED BENCHMARK TIMINGS ARE UNREPRODUCED. The prior agent reported running things locally: the Idukki-scale MPC port (45 solves in 0.39 s), the rtc-tools single_reservoir example (3.07 s), the grid-vs-NSGA-II head-to-head (2.02 s vs 2.31 s on 4,800 evals), the pywr install (37 packages, 30 s) and the CasADi bundled-plugin enumeration. I checked this environment's Python: casadi,…
- RL NEGATIVE FINDING SURVIVES, with two corrections. I re-ran six GitHub searches adversarially. 'reservoir+operation+DDPG' does return 0 repositories, exactly as claimed. But broader searches surface things the prior agent missed. The top hit by stars for 'reservoir + reinforcement learning' is Muzhaffar99/RL-Based-Control-of-Reservoir-Systems-Using-SAC-and-PPO at 38 stars - I checked it and it…
- LICENCE CORRECTIONS the GitHub API gets wrong, all of which matter for a KSEB-bound deliverable. MOEAFramework reports NOASSERTION only because its licence sits in a file named COPYING; I read it and it is LGPL-3.0-or-later. SDDP.jl also reports NOASSERTION; its LICENSE.md states plainly 'SDDP.jl is licensed under the Mozilla Public License, Version 2.0' - permissive and safe to build on.…
- OVERSTATED BY DEGREE, not fabricated. (a) Lake Mendocino: 'FIRO is now authorised in the USACE Water Control Manual' overstates what the Dec 2020 FVA itself says - that document RECOMMENDS a five-year major deviation and a WCM update; at the time of writing FIRO ran through major deviations. The '1959, updated only twice' framing IS supported (the PDF gives 'Original 1959, Updated 1986, Minor…
- UNVERIFIABLE, PLUS BOT-BLOCKED ITEMS CONFIRMED BY OTHER MEANS. HEC-ResSim 'v4.1, April 2026' could NOT be checked - both hec.usace.army.mil URLs returned status 000 (connection failure; the .mil host is unreachable from here, not merely bot-blocked). The surrounding argument is consistent with what is known about HEC-ResSim, but drop the version specificity. Per the brief's instruction on…

- WHAT HELD UP BEST, for calibration of how much to trust the rest. The two verdicts I tried hardest to break both survived. (1) RTC-Tools' Modelica lock-in: I scripted a per-example check across all 15 examples - every one of the 13 optimisation examples imports ModelicaMixin, and the only two that do not are simulation examples that still ship a .mo file. 'Do not port, steal the two designs' is…

Rejected (5)

Recorded so the same ground is not covered twice.

Claimed source	Why it was rejected
Borg MOEA project site	URL is live (200) and the prior agent's licensing warning is CORRECT and now double-confirmed - but that is precisely why it must not appear on a list of OPEN-SOURCE sources. borgmoea.org states verbatim: 'Borg is…
bernardoct/WaterPaths	GitHub repo exists (23 stars, Apache-2.0, verified via API) but fails on both relevance and liveness. Last pushed 2022-01-04 - dead for over four and a half years. More importantly it solves a different problem:…
FeralFlows/pysedsim	GitHub repo exists (18 stars, BSD-2-Clause, verified via API) but last pushed 2020-07-22 - dead for six years. Its distinguishing feature is SEDIMENT simulation for dam siting and sediment-management design, which…
Critical-Infrastructure-Systems-Lab/VICResOpt	GitHub repo exists (35 stars) and the NSGA-II driver claim is TRUE - I confirmed Routing/OptCalib/autocalieNSGAII.py, multicalieNSGAII.py and optimization.py. Rejected anyway on licensing: the GitHub API reports no…
pywr/pywr-next (Pywr 2.0, Rust)	Every fact the prior agent reported is confirmed exactly - 12 stars, 54 open issues, created 2020-11-17, description still verbatim 'An experimental repository exploring ideas for a major revision to Pywr using Rust as…

7 · Open Source River Routing, Hydraulics And 2D Flood Inundation (Adversarial Verification Pass)

Confirmed sources (30)

Source	Kind	What it provides	Relevance	HTTP
CWC River Water Level (Manual - Hourly), basin 031 - India National Water Data Portal data throu · page does not print a licence string in the fetched view; portal-wide terms are 'Other (Open)' https://nwdp.nwic.gov.in/dataset/d951a09c-6cf8-470e-be77-…	government open data (CKAN, no login)	Hourly manual river stage + reservoir level CSVs by BASIN code. I re-downloaded nothing but independently re-parsed both cached files with my own script: wl031.csv = 1,284,651 rows spanning… caveat: THREE material corrections. (1) ARANGALI IS NOT ON THE PERIYAR. The file's own metadata says Tributary=Chalakudy, Local River=Chalakudy,…	HIGHEST. Genuinely closes gap #1 (no pre-2020 Periyar data) for river stage. Provides an…	200
CWC River Discharge (Manual - Daily), Kerala data throu https://nwdp.nwic.gov.in/dataset/river-discharge-manual-d…	government open data (CKAN, no login)	Daily gauged discharge. Independently re-parsed the cached 2001-2025 Kerala file: 217,538 rows, 40 distinct stations - both EXACT matches. NEELEESWARAM (River=Periyar, Local River=Periyar) has 8,478… caveat: The 7-day hole is REAL and I confirmed it precisely: 2018-08-16 through 2018-08-22 are all absent, next value is 2018-08-23 = 924.282.…	HIGH. The discharge half of the rating-curve route, and an independent check on the…	200
CWC Study Report: Kerala Floods of August 2018 (Hydrology (S) Dte) Government of India public report https://sdma.kerala.gov.in/wp-content/uploads/2020/08/CWC…	authoritative government study report (PDF)	I read the extracted text directly. Confirmed VERBATIM: contents list has '4.1 Runoff computations of Periyar sub-basin', '4.1.1 Reservoir operation of Idukki', '4.1.2 Reservoir operation of… caveat: Bonus cross-validation the prior sweep missed: the report states Idukki gross storage 1997 MCM at FRL 732.43 m, MDDL 694.94 m, and 'live…	HIGHEST for the 2018 gap. Supplies the dam-outflow boundary condition that the CWC gauge…	200
Deltares/SFINCS (+ hydromt_sfincs) SFINCS 83, hydromt_sfincs 35★ · SFINCS pus · GPL-3.0 (source); Deltares Freeware (binaries) https://github.com/Deltares/SFINCS	GitHub repo (Fortran solver) + Python model builder	Reduced-complexity 2D compound-flood solver. SFINCS docs confirm the variant naming VERBATIM: 'Traditionally SFINCS neglects the advection term (SFINCS-LIE) which generally justified for… caveat: Licence is split and the prior sweep glossed it: SOURCE is GPL-3.0, but the precompiled Windows executable ships under the Deltares…	HIGHEST for gap #4 (2D inundation). Only candidate needing no Fortran compiler and no…	200
NOAA-OWP/t-route 55★ · pushed_at · Apache-2.0 https://github.com/NOAA-OWP/t-route	GitHub repo (Python over Fortran kernels)	Muskingum-Cunge and diffusive-wave routing. README confirms the parallelism split verbatim: MC is 'Parallel on CONUS using junction order and short-time step', diffusive is 'Serial on a sub-network… caveat: The staleness call checks out and is if anything understated. GitHub pushed_at is 2026-08-18 but the last commit ON master is 2025-01-31…	LOW-MEDIUM. AquaSync already has Muskingum with sub-reaching. The prior sweep's own 'not…	200

Source	Kind	What it provides	Relevance	HTTP
rileyhales/river-route 4★ · PyPI 2.1.1 · BSD-3-Clause-Clear https://github.com/rileyhales/river-route	GitHub repo + PyPI package	Vectorised Muskingum-family routing on vector stream networks. Three routers confirmed: Muskingum ('Channel routing only (no lateral inflow)'), RapidMuskingum, UnitMuskingum. Uses 'numba-compiled… caveat: No calibration module, confirmed: the docs say k and x come 'from hydraulic assumptions and/or calibration' performed elsewhere. Only 4…	MEDIUM. Good short reference implementation of sparse-matrix Muskingum; not a calibrator.	200
LISFLOOD-FP 8.x (SEAMLESS-WAVE / Sheffield) https://www.seamlesswave.com/LISFLOOD8.0.html	academic model distribution page	Canonical academic 2D floodplain code. Page confirms DG2/FV1 solvers, GPU-accelerated ACC solvers, versions 8.1 and 8.2, a GPU multiwavelet DG2 solver with dynamic adaptivity (GPU-MWDG2), a DG2-RANS… caveat: I chased the citation because it looked hallucination-shaped and it is NOT: https://gmd.copernicus.org/articles/18/9827/2025/ resolves 200…	MEDIUM. Heavier route to gap #4 than SFINCS; the prior sweep's 2-4 weeks vs 2-4 days…	200
Deltares/pyflwdir 116★ · pushed 202 · MIT https://github.com/Deltares/pyflwdir	GitHub repo + PyPI package	Pure-Python numba-accelerated terrain/hydrography analysis. Docs list confirmed VERBATIM: 'flow directions from elevation data using a steepest gradient algorithm / strahler stream order / flow…	HIGH. The practical way to get HAND and a conditioned DEM for the Aluva reach without a…	200
Deltares/hydromt_sfincs 35★ · pushed 202 · GPL-3.0 https://github.com/Deltares/hydromt_sfincs	GitHub repo + PyPI package	Python plugin that automates SFINCS model building, described as tools 'to go from raw data to a complete model instance which is ready to run and to analyse model results'. Usable as a CLI driven…	HIGH. This is what makes the SFINCS route a 2-4 day job rather than a schematisation slog.	200
gpt-cmdr/ras-commander 76★ · pushed 202 · MIT https://github.com/gpt-cmdr/ras-commander	GitHub repo + PyPI package	Python API for HEC-RAS 6.x. Confirmed: 'Parallel Execution of Multiple Plans' via RasCmdr.compute_parallel(); standalone HDF result reading without HEC-RAS installed; an HdfResultsMesh class doing… caveat: Modern HDF features target HEC-RAS 6.2+; legacy COM support covers 3.1, 4.1, 5.0.x, 6.0, 6.3, 6.6. The prior sweep's claim that pyHMT2D is…	MEDIUM. The HDF reader is genuinely useful standalone; the execution half depends on…	200
psu-efd/pyHMT2D 127★ · pushed 202 · MIT https://github.com/psu-efd/pyHMT2D	GitHub repo + PyPI package	Control and semi-automation of 2D hydraulic models with pre/post-processing: edit Manning's n and boundary conditions, run 'batch simulations to calibrate model runs', Monte Carlo uncertainty… caveat: Requires the external software (SRH-2D via SMS, and/or HEC-RAS) to be installed separately - it is a driver, not a solver. Given HEC-RAS…	MEDIUM. The batch-calibration and Monte Carlo machinery is the interesting part for…	200
Landlab (OverlandFlow component) 437★ · pushed 202 · MIT https://landlab.csdms.io/	Python modelling framework	OverlandFlow implements the de Almeida et al. (2012) local-inertial approximation of the 2D shallow water equations on a raster grid. Component citation confirmed: Adams et al. (2017), 'The Landlab…	MEDIUM. A pip-installable 2D local-inertial solver - the same physics family as…	200

Source	Kind	What it provides	Relevance	HTTP
MERIT Hydro / MERIT DEM CC BY-NC 4.0 or ODbL 1.0 (dual) https://global-hydrodynamics.github.io/MERIT_Hydro/	global raster hydrography dataset	90 m (3 arc-second) globally consistent hydrography on the error-corrected MERIT DEM, 90N-60S, 5x5 degree GeoTIFF tiles. Six layers confirmed: flow direction, upstream drainage area, upstream… caveat: LICENCE OMITTED BY THE PRIOR SWEEP and it is restrictive: dual-licensed CC BY-NC 4.0 (non-commercial) OR ODbL 1.0 (commercial use allowed…	HIGH. The conditioned-DEM and HAND base layer for any 2D or HAND-based work on the…	**200**
MERIT-Basins (reachhydro.org) https://www.reachhydro.org/home/params/merit-basins	global vector hydrography dataset	'A global vector hydrography database derived from the 90-m MERIT-Hydro product using the TauDEM tool.' ~2.94M river reaches plus unit catchments. File naming confirmed:…	HIGH. This is the object that settles the Periyar/Chalakudy reach topology and supplies…	**200**
GRFR - Global Reach-level Flood Reanalysis https://www.reachhydro.org/home/records/grfr	global modelled discharge reanalysis	VIC land-surface model coupled to RAPID routing. 3-hourly and daily river discharge for the same 2.94M MERIT-Basins reaches, 1980-2019. Flood events extracted above a 2-year return threshold.…	MEDIUM-HIGH. A modelled 1980-2019 discharge series for Periyar reaches - covers 2018 -…	**200**
delta-HBV2 / delta-MC2 global streamflow (Nature Communications 2025 + Zenodo) dataset v5 · PSU Non-Commercial Software License Agreement https://www.nature.com/articles/s41467-025-64367-1	journal paper + companion dataset	Paper: 'Distinct hydrologic response patterns and trends worldwide revealed by physics-embedded learning', Ji, Song, Bindas, Shen, Yang, Pan, Liu, Rahmani, Abbas, Beck, Lawson & Wada, Nature… caveat: Licence is a Pennsylvania State University Non-Commercial Software License Agreement; commercial use needs 'prior written consent from…	MEDIUM. Gives you a differentiable-MC-routed discharge series over the Periyar for…	**200**
NOAA-OWP/inundation-mapping (HAND-FIM) 129★ · pushed 202 · Apache-2.0 https://github.com/NOAA-OWP/inundation-mapping	GitHub repo (operational framework)	Confirmed verbatim: 'This software uses the Height Above Nearest Drainage (HAND) method to generate Relative Elevation Models (REMs), Synthetic Rating Curves (SRCs), and catchment grids', producing… caveat: IMPORTANT OMISSION IN THE ORIGINAL WRITE-UP: this is hard-wired to US infrastructure. The README opens by saying it is 'configured to work…	MEDIUM as a METHOD reference for gap #4; LOW as a drop-in tool, for the reason below.	**200**
Copernicus DEM GLO-30 (AWS Open Data) free for the general public under Copernicus DEM terms https://registry.opendata.aws/copernicus-dem/	global DEM, cloud-hosted	'A Digital Surface Model (DSM) which represents the surface of the Earth including buildings, infrastructure and vegetation.' GLO-30 Public at 30 m, GLO-90 at 90 m worldwide. Anonymous S3 access… caveat: Two things to weigh. (1) GLO-30 Public is noted as having 'limited worldwide coverage' - verify the Periyar tiles exist before planning…	HIGH. The licence-clean DEM baseline for a SFINCS Aluva/Kochi model - and unlike FABDEM…	**200**

Source	Kind	What it provides	Relevance	HTTP
FABDEM V1-2 (Forest And Buildings removed Copernicus DEM) CC BY-NC-SA 4.0 (NON-COMMERCIAL) https://research-information.bris.ac.uk/en/datasets/fabde…	global bare-earth DEM	'FABDEM (Forest And Buildings removed Copernicus DEM) is a global elevation map that removes building and tree height biases from the Copernicus GLO 30 Digital Elevation Model (DEM).' 1 arc-second… caveat: LICENCE IS CC BY-NC-SA 4.0 - NON-COMMERCIAL AND SHARE-ALIKE. Quote: 'The FABDEM dataset is licensed under a Creative Commons CC BY-NC-SA…	HIGH technically - bare-earth is what a floodplain model wants - but see the licence…	200
GUARDIAN - sub-daily river discharge over India https://pmc.ncbi.nlm.nih.gov/articles/PMC11489425/	journal paper (Scientific Data 2024) + method precedent	'ExtendinG SUb-DAily River Discharge data over INdia (GUARDIAN)', Scientific Data, 18 Oct 2024. Extracts water-surface-elevation from 1,365 India-WRIS stations, fits rating curves with four… caveat: The paper does NOT specifically mention Kerala or the Periyar, so do not assume a ready-made Periyar rating curve exists in it. Its value…	HIGH as METHOD VALIDATION. This is peer-reviewed precedent for exactly the…	200
GHI / GHSA - Geospatial dataset for Hydrologic analyses in India CC BY 4.0 https://essd.copernicus.org/articles/15/4389/2023/	journal paper (ESSD 2023) + Zenodo dataset	Quality-controlled CWC/WRIS gauge metadata, catchment boundaries and hydrometeorological time series. Zenodo 13852439 confirmed: 'Geospatial dataset for hydrologic analyses within the Indian… caveat: Licence is CC BY 4.0 (permissive - better than FABDEM or MERIT Hydro for this project). The Zenodo page does not state the year range, so…	MEDIUM-HIGH. Independent, vetted catchment boundaries for the Periyar gauges - useful…	200
Global Flood Database v1 (2000-2018) https://developers.google.com/earth-engine/datasets/catal…	Earth Engine image collection	'Maps for 913 flood events from 2000-2018', MODIS Terra/Aqua at 250 m, availability 2000-02-17 to 2018-12-10. Bands: flooded (maximum extent), duration, clear_views, clear_perc, jrc_perm_water. 'Can… caveat: Aug 2018 falls inside the collection's window (ends 2018-12-10), so the claim is plausible, but confirm by filtering the collection on…	MEDIUM-HIGH. The only ready-made observed inundation EXTENT for validating a 2D Aug-2018…	200
anuga-community/anuga_core (ANUGA) 53 (anuga-community) / 227 (GeoscienceAustralia upstream)★ · pushed 202 · NOASSERTION per GitHub API - verify in-repo https://anuga.readthedocs.io/	GitHub repo + PyPI package	Finite-volume unstructured-triangle 2D shallow water solver for dam-break, riverine and tsunami inundation; Python with C kernels. caveat: REPO PROVENANCE: anuga-community/anuga_core is flagged fork=true by the GitHub API (parent stoiver/anuga_core, ultimate source…	MEDIUM. A genuine open-source alternative to SFINCS with true unstructured meshes;…	200

Source	Kind	What it provides	Relevance	HTTP
open TELEMAC-MASCARET https://www.opentelemac.org/index.php/download	CFD suite (EDF-led consortium)	TELEMAC-2D solves the depth-averaged Saint-Venant equations on unstructured meshes; consortium-governed open source. caveat: The cert-chain defect means the specific version and licence claims are UNVERIFIED. Do not quote a TELEMAC version number without checking…	LOW-MEDIUM for AquaSync. Powerful but by far the heaviest lift of the 2D options; hard…	200
HydroSHEDS / HydroRIVERS https://www.hydrosheds.org/products/hydrorivers	global vector river network	Global river network line shapefiles at 15 arc-second resolution, distributed by continent, with HydroBASINS catchments and HydroATLAS attributes alongside. caveat: Lower-confidence entry: status-checked only, not content-audited. Treat the 15 arc-second figure as unconfirmed by me.	MEDIUM. A second, independent vector network to cross-check MERIT-Basins topology on the…	200
jsosa/LFPtools 40★ · pushed 202 · BSD-3-Clause per GitHub API, MIT per setup.py - inconsistent https://github.com/jsosa/LFPtools	GitHub repo (Python CLI suite)	CLI tools to prepare LISFLOOD-FP inputs from global data. I listed the repo's bin/ directory and confirmed every claimed tool exists plus more: lfp-getdepths, lfp-getwidths, lfp-getbankelevs,… caveat: EFFECTIVELY ABANDONED: last push 2021-04-19, over five years ago. Also a licence contradiction - the GitHub API reports BSD-3-Clause while…	LOW. Only useful if you commit to LISFLOOD-FP over SFINCS, and the staleness makes that…	200
mhpi/dMC-Juniata-hydroDL2 (differentiable Muskingum-Cunge) 11★ · pushed 202 · CC-BY-4.0 https://github.com/mhpi/dMC-Juniata-hydroDL2	GitHub repo (paper replication code)	PyTorch + Hydra implementation of Bindas et al. 2024 WRR: Muskingum-Cunge with an embedded neural net mapping reach attributes to Manning's n and channel geometry, trained on downstream hydrographs… caveat: Confirmed as paper-replication rather than a library: 11 stars, last push 2024-04-14 (over two years), ZERO open issues and zero PRs, 6…	MEDIUM as the conceptual answer to gap #2 - learning routing parameters from downstream…	200
DeepGroundwater/ddr (Distributed Differentiable Routing) 11★ · pushed 202 · Apache-2.0 https://github.com/DeepGroundwater/ddr	GitHub repo	The actively maintained successor line to dMC-Juniata for differentiable routing. caveat: Small project (11 stars). I confirmed its metadata but did not audit its README or API surface, so treat capability claims as unverified.	MEDIUM. If AquaSync pursues learned routing parameters, this is the maintained codebase…	200
ec-jrc/lisflood-code (LISFLOOD-OS) 167★ · pushed 202 · EUPL-1.2 https://github.com/ec-jrc/lisflood-code	GitHub repo (distributed hydrological model)	Spatially distributed rainfall-runoff-routing model in Python, requiring PCRaster and GDAL. Repo topics include 'efas', 'glofas', 'hydrological-modelling' and 'flood-forecasting'. caveat: The reservoir/dam representation - the one part that might interest AquaSync - is precisely the part I could not confirm from the repo…	LOW for this project. AquaSync already has SCS-CN rainfall-runoff and Muskingum routing;…	200

Source	Kind	What it provides	Relevance	HTTP
ISRO Bhoonidhi (NRSC) - registration portal https://bhoonidhi.nrsc.gov.in/bhoonidhi/registration.html	national EO data portal (registration required)	Confirmed to be the genuine NRSC/ISRO Bhoonidhi registration page. Form collects name, user ID, password, country, user type (Academic / Private / Central Government / State government / DOS NRSC)… caveat: DO NOT PLAN AROUND THE CARTODEM CLAIMS until verified. Register and inspect the actual catalogue before treating CartoDEM 30 m as an…	POTENTIALLY HIGH but UNPROVEN. An Indian national DEM would be the natural DEM for an…	200

Verification notes

- BIGGEST CORRECTION - THE MUSKINGUM HEADLINE IS WRONG. The claim that 'two gauges on one river with synchronous hourly stage is exactly the input a Muskingum K/x fit needs' does not survive. The CWC file's OWN metadata settles it: ARANGALI has Tributary=Chalakudy, Local River=Chalakudy, District=THRISSUR. It is on the Chalakudy, which joins the Periyar DOWNSTREAM of Neeleeswaram - so the two…
- THE SALVAGE, AND IT IS STILL GOOD. There IS a defensible Muskingum reach here, just not the claimed one: dam outflow (Idukki spill + powerhouse, plus Idamalayar spill) routed down to NEELEESWARAM. That is precisely the reach AquaSync cares about (dams to Aluva). For August 2018 the upstream boundary comes from the CWC report's hourly Idukki/Idamalayar operation narrative and Fig.4, and the…
- THE UNDERLYING DATA CLAIMS ARE EXCEPTIONALLY SOLID - I could not break them. I re-parsed both cached CSVs with my own script and every row count reproduced EXACTLY: 1,284,651 and 1,139,984 total; NEELEESWARAM 31,172+25,643; ARANGALI 30,186+27,460; VANDIPERIYAR 48,949+26,714; Idukki 3,532+9,477; Idamalayar 3,621+9,561. Individual values matched too: Neeleeswaram 6.12 m at 2018-08-14 00:00 and…
- THE CORRUPT-WINDOW REPAIR CLAIM IS OVERSTATED BY ROUGHLY HALF. 'Spans the corrupt 2020-09-25 -> 2021-04-30 window' is not true. The reservoir series runs 2019-06-10 to 2020-12-31 and then JUMPS to 2021-05-12 - a four-and-a-half month hole. It supplies 881 (Idukki) / 882 (Idamalayar) readings inside the corrupt window, all of them at or before 2020-12-31 22:00, and NOTHING for 2021-01-01 to…
- STAGE COVERAGE IS SEASONAL, NOT CONTINUOUS HOURLY - which is fine, but plan for it. NEELEESWARAM 2018 monthly completeness measured against unique timestamps: Aug 100%, Jul 98%, Oct 89%, Sep 87%, Jun 84%, Nov 69%, but only 11-15% for Dec-May. So '31,172 hourly stage readings 2015-06-01 to 2020-12-31' is literally true but reads as continuous when it is monsoon-hourly and dry-season-sparse.…
- UNDOCUMENTED DATA-QUALITY LANDMINE: DUPLICATE AND CONFLICTING TIMESTAMPS. In the 1991-2020 file, NEELEESWARAM has 31,172 rows but only 29,835 unique timestamps - 1,337 duplicate rows, and 667 of those timestamps carry CONFLICTING values (e.g. 2018-08-11 08:00 appears as both 6.84 and 6.80). The 2021-2025 file is clean (zero duplicates). Any least-squares K/x fit or rating-curve regression must…
- LICENCE RESTRICTIONS WERE OMITTED ACROSS THE BOARD, AND THEY MATTER FOR A KSEB/KSDMA HANDOFF. FABDEM is CC BY-NC-SA 4.0 - explicitly NON-COMMERCIAL and share-alike. MERIT Hydro is dual CC BY-NC 4.0 / ODbL 1.0, where the commercial option forces derived data to be published under ODbL share-alike. The delta-HBV2 Zenodo dataset carries a Penn State Non-Commercial Software License needing written…

- THE US-LOCK-IN OBJECTION WAS APPLIED TO t-route BUT NOT TO inundation-mapping, THOUGH IT APPLIES EQUALLY. The t-route assessment is accurate and I confirmed it verbatim ('bespoke for NWM', WSL required on Windows, gfortran + netcdf-fortran + compiler.sh, master frozen since 2025-01-31 despite a 2026-08-18 pushed_at). But NOAA-OWP/inundation-mapping is described as an operational…
- TWO CITATIONS I EXPECTED TO BE FABRICATED TURNED OUT REAL - worth recording so they are not needlessly dropped. (1) 'GMD 18, 9827, 2025' for LISFLOOD-FP 8.2 resolves 200 and is genuinely Chowdhury & Kesserwani, 'LISFLOOD-FP 8.2: GPU-accelerated multiwavelet discontinuous Galerkin solver with dynamic resolution adaptivity', GMD 18(23), 9827-9854, 10 Dec 2025 - even though the seamlesswave page…
- TOOLING VERDICT AFTER ADVERSARIAL CHECK: the SFINCS recommendation survives and is the right call for gap #4. SFINCS-LIE/SFINCS-SSWE naming, GPL-3.0 source, the Windows exe at download.deltares.nl/sfincs, the deltares/sfincs-cpu Docker image, release v2.4.0_Galibier 2026-06-15, and hydromt_sfincs (GPL-3.0, 35 stars, PyPI 1.2.2 on 2025-12-02) all confirmed. Pair it with pyflwdir (MIT, 116 stars,…

Rejected (2)

Recorded so the same ground is not covered twice.

Claimed source	Why it was rejected
HEC-RAS (USACE) downloads	UNVERIFIABLE FROM HERE, AND THE VERSION CLAIMS ARE HALLUCINATION-SHAPED. I retried three URLs on the domain (the download page, the HEC-RAS product page, and the site root) and every one returned 000 - the entire…
Iber (+ IberSWMM+)	NOT OPEN SOURCE - fails the dimension - AND TWO KEY CLAIMS UNCONFIRMED. The URL resolves 200 and Iber is a real, capable 2D model, but its own download page states the licence terms as 'The reproduction,…

8 · State Of The Art In Dam Operation Decision Support: What Is Deployed, What Fails

Confirmed sources (30)

Source	Kind	What it provides	Relevance	HTTP
CAG Report No. 6 of 2021 - Performance Audit: Preparedness and Response to Floods in Kerala Presented · Government of India / CAG - public audit report tabled in the State Legislature https://cag.gov.in/uploads/download_audit_report/2021/Ker…	Official government performance audit (PDF, 144 pages, 6.68 MB, tabled 11 Nov 2021)	Downloaded and parsed page by page. VERIFIED VERBATIM: IISc Bangalore engaged as Consultant, Prof. P. Pradeep Mujumdar and team (pdf p.21, p.48). Idukki spills 14-18 Aug 2018: actual 467.51 MCM vs… caveat: Do NOT ingest the level/storage/spill columns of Appendices 3.5-3.7 as observed 2018 data. Only the inflow column (and PH discharge /…	Single most valuable document for AquaSync. Supplies the exact official 2020 rule curve…	200
CWC Study Report: Kerala Floods of August 2018 (Hydrology (S) Directorate, September 2018) Report dat · Government of India / CWC, hosted by Kerala SDMA https://sdma.kerala.gov.in/wp-content/uploads/2020/08/CWC…	Official Central Water Commission study report (PDF, 48 pages, 6.0 MB)	Downloaded and parsed. VERIFIED: Idukki 731.82 m at 00:00 on 10 Aug, 0.61 m below FRL, ~40 MCM cushion (p.12); drawn to 730.80 m by 13 Aug giving ~90 MCM (p.13); peak inflow 2532 cumec at 22:00 on…	The primary official 'dams did not cause it' analysis and the counterweight to CAG/IISc.…	200
Sudheer et al. (2019), Role of dams on the floods of August 2018 in Periyar River Basin, Kerala, Current Science 116(5):780-794 Published · Current Science / Indian Academy of Sciences - freely downloadable PDF https://www.currentscience.ac.in/Volumes/116/05/0780.pdf	Peer-reviewed research article (PDF, 15 pages), doi 10.18520/cs/v116/i5/780-794	Downloaded and parsed. THE HEADLINE FORECAST-SKILL CLAIM IS CONFIRMED VERBATIM (p.789): 'The probability of detection (POD) of higher rainfall (>25 mm/day) at short range (3-day) or medium range…	The strongest single argument against a perfect-foresight optimiser in this basin, and…	200
Mishra et al., The Kerala flood of 2018: combined impact of extreme rainfall and reservoir storage (HESS Discussions preprint hess-2018-480) Discussion · CC BY 4.0 https://hess.copernicus.org/preprints/hess-2018-480/	Open discussion preprint with full public review thread, doi 10.5194/hess-2018-480	Fetched and parsed the live page. STATUS LINE CONFIRMED VERBATIM: 'this discussion paper is a preprint. It has been under review for the journal Hydrology and Earth System Sciences (HESS). The…	Shows that the most-quoted 'reservoirs made it worse' study was never peer-reviewed.…	200
SANDRP guest article: Role of dams in Kerala floods - Distortion of Science (J. Harsha, Director, CWC Chennai) Published · SANDRP - freely readable; views stated as personal https://sandrp.in/2020/08/25/role-of-dams-in-kerala-flood…	Technical rebuttal published as a guest article after journal refusal	Fetched and parsed (241 KB, 78 comment elements). VERIFIED: attacks Sudheer et al.'s stated assumptions, listed explicitly as 'SCS-CN method and lambda = 0.2 is valid in PRB' and 'The land use/land…	Harsha's two charges land directly on AquaSync's own method: an SCS-CN with lambda=0.2…	200

Source	Kind	What it provides	Relevance	HTTP
SANDRP, the two halves of the Sudheer/Harsha exchange (19 Sep 2020) - response and rejoinder Both publi · SANDRP https://sandrp.in/2020/09/19/response-of-sudheer-et-al-to…	Two separate SANDRP articles published the same day - both fetched and verified 200	PART 1 (the URL above, Sudheer et al.'s response): contains the IIT Madras defence of lambda=0.2 verbatim - 'in the case of Sudheer et al (2019), where simulation of high flows are performed with…	Part 1 is the technically strongest available defence of exactly the modelling choice…	200
Vegad & Mishra (2025), Climate change and effectiveness of dams in flood mitigation in India, npj Natural Hazards 2:63 Published · CC BY-NC-ND 4.0 https://www.nature.com/articles/s44304-025-00117-z	Peer-reviewed open-access article, doi 10.1038/s44304-025-00117-z (10-page PDF downloaded)	Downloaded the OA PDF and parsed. VERIFIED: 178 major dams, in-situ and satellite observations plus H08/CaMa-Flood model simulations; 'flood mitigation depends more on antecedent reservoir storage… caveat: The standard deviations exceed the medians (4.7 +/- 6.7, 14.6 +/- 15.11) - the projected signal is noisy across GCMs.	Independent, India-wide, recent confirmation that antecedent storage - not forecast…	200
Suresh et al. (2024), Satellite-based Tracking of Reservoir Operations for Flood Management during the 2018 Extreme Weather Event in Kerala, India - open NASA NTRS manuscript Manuscript · NASA NTRS public access copy of the accepted manuscript https://ntrs.nasa.gov/api/citations/20240007715/downloads…	Accepted-manuscript PDF (56 pages) of the Remote Sensing of Environment paper, doi 10.1016/j.rse.2024.114149, RSE Vol. 307, June 2024	Downloaded and parsed. Table 4 confirmed EXACTLY as claimed: Idamalayaar r=0.81 / RMSE 135.29 cumec / NRMSE 13.30%; Banasurasagar 0.78 / 31.81 / 14.14%; IDUKKI 0.63 / 217.94 cumec / 12.45%; Kakki…	The published skill ceiling for satellite-derived Idukki inflow (r=0.63) is a hard…	200
Das et al. (2025), Forecast-Informed Reservoir Operations within a Satellite-Based Framework for Mountainous and High-Precipitation Regions: Case of the 2018 Kerala Floods, J. Hydrologic Engineering 30(2) April 2025 · Closed access - OpenAlex reports is_oa = false, no open version anywhere https://ascelibrary.org/doi/10.1061/JHYEFF.HEENG-6276	Peer-reviewed journal article, doi 10.1061/JHYEFF.HEENG-6276	Existence and metadata confirmed independently via Crossref: exact title as claimed, Journal of Hydrologic Engineering vol. 30 issue 2, issued April 2025, ASCE, authors Das P; Suresh S; Hossain F;… caveat: Paywalled with no green OA copy. This is the one paper on the list that is both closest to AquaSync's thesis and genuinely unreadable -…	Closest published prior art to AquaSync's core thesis: forecast-informed reservoir…	403-bot-blocked (ASCE blocks curl and WebFetch alike; existence and full metadata confirmed via api.crossref.org)

Source	Kind	What it provides	Relevance	HTTP
KSDMA Dam Water Level portal (daily KSEB and Irrigation bulletins) Live, upda · Government of Kerala / KSDMA https://sdma.kerala.gov.in/dam-water-level/	Official state disaster-management authority live data page	Fetched and parsed today. Live and current: the page now carries 'Water Levels of Major Reservoirs (KSEB) - 26/08/2026 - 11.00 AM' and 'Water Levels of Major Reservoirs (IRRIGATION) - 26/08/2026 -…	The stable, official entry point. Better provenance than the GitHub scrape AquaSync…	200
KSDMA daily KSEB dam bulletin (current file) File last · KSEB Ltd / KSDMA, public bulletin https://sdma.kerala.gov.in/wp-content/uploads/2026/08/KSE…	Daily operational bulletin PDF (1 page, 65 KB, produced from DAILY ALERTS 2026.xlsx via Print To PDF)	Downloaded and parsed. Header dated 25/08/2026 - 11.00 AM, PDF creationDate D:20260825094020+05'30'. Schema confirmed as 14 columns: Sl.No, Name of the Dam, District, FRL, Today's Water Level, rule… caveat: This URL is NOT a stable archive endpoint. The portal still points at KSEB-SITE-13.pdf on 26/08 while the file itself still holds the…	Confirms AquaSync's 1459 Mm3 Idukki live storage against the official figure (1459.49…	200
KSEB Limited Dam Safety Organisation Live; FRL · CC BY-SA 2.5 IN (stated in the site footer) https://dams.kseb.in/?page_id=39	Official dam-owner institutional site	Fetched and parsed. Confirmed table 'Basic Details of 59 Dams of KSEB Limited' with columns Name, District, FRL (m), gated yes/no, large dam yes/no, criteria for major-dam classification, including…	The institutional counterpart AquaSync must integrate with under the Dam Safety Act, and…	200
Dam Safety Act 2021 (Act No. 41 of 2021) - full gazette text Act dated · Government of India statute - public domain https://wrd.maharashtra.gov.in/Site/Upload/Pdf/DSA_2021_a…	Full statute text, gazette reprint (PDF, 21 pages, 459 KB)	Downloaded and parsed. Header confirmed: 'THE DAM SAFETY ACT, 2021 / NO. 41 OF 2021 / [13th December, 2021.]' with 'THE GAZETTE OF INDIA EXTRAORDINARY [PART II-' running footers. VERIFIED VERBATIM -… caveat: This is a state-government (Maharashtra WRD) mirror of the central gazette, not the canonical India Code or eGazette copy. The text is a…	India already has a statutory mandate for precisely what AquaSync does (s.35(1)(a)…	200
Lake Mendocino FIRO Final Viability Assessment (December 2020) December 2 · Public - also archived at eScholarship (escholarship.org/uc/item/3b63q04n) https://cw3e.ucsd.edu/FIRO_docs/LakeMendocino_FIRO_FVA.pdf	Steering-committee viability assessment (PDF, 141 pages, 11.3 MB)	Downloaded and parsed. VERIFIED: Table E.1 evaluates four FIRO options plus current operations against 16 objective metrics, with median 10-May storage increase over baseline of 0% (baseline), 27%…	The definitive FIRO reference, and the source of the honest forecast-skill decay numbers…	200
Prado Dam FIRO Final Viability Assessment (December 2023) December 2 · Public (CW3E) https://cw3e.ucsd.edu/FIRO_docs/Prado/FIRO_Prado_FVA.pdf	Steering-committee viability assessment (PDF, 136 pages, 10.0 MB)	Downloaded and parsed. VERIFIED: 'In April 2021, the Prado Dam interim WCM was modified to increase the buffer pool elevation from 498 feet to 505 feet throughout the year', the current buffer pool…	Second completed FVA; shows the same deviation-first, manual-last institutional route as…	200

Source	Kind	What it provides	Relevance	HTTP
Scripps/UCSD: New forecast-informed decision-making tool implemented at Coyote Valley Dam and Lake Mendocino 22 October · UC San Diego news, freely readable https://scripps.ucsd.edu/news/new-forecast-informed-decis…	Institutional news release, dated Oct 22, 2025	Fetched and parsed. VERIFIED VERBATIM: signing ceremony 22 October 2025 by USACE, Sonoma County Water Agency, UC San Diego Scripps CW3E and California DWR; 'The revised manual redefines the…	Closes the FIRO institutional timeline: 2014 steering committee to signed water control…	200
USACE/Sonoma Water Draft EA: Water Year 2021-2026 Major Planned Deviation to the Coyote Valley Dam-Lake Mendocino Water Control Manual October 20 · US Government NEPA document, public https://www.sonomawater.org/media/PDF/Environment/Environ…	NEPA Draft Environmental Assessment (PDF, 127 pages, 17.2 MB), OCTOBER 2020	Downloaded and parsed. Cover verified: 'Water Year 2021 - 2026 Major Planned Deviation to the Coyote Valley Dam-Lake Mendocino Water Control Manual, Draft Environmental Assessment, OCTOBER 2020',…	The NEPA instrument that bridged the 2020 FVA and the 2025 permanent WCM, and the proof…	200
Forbis & Ly (2025), Application of forecast-informed reservoir operations at US Army Corps of Engineers dams in California, J. Flood Risk Management 18(1) Online 16 · CC BY 4.0 (gold OA) https://doi.org/10.1111/jfr3.13051	Peer-reviewed open-access article (CC BY 4.0), doi 10.1111/jfr3.13051	Metadata and abstract confirmed via Crossref and the DOAJ API. Authors are Joe Forbis (USACE ERDC Coastal and Hydraulics Laboratory, Sacramento) and Cuong Ly (USACE South Pacific Division, Los…	USACE-authored account of what actually blocks FIRO deployment. The abstract…	**403-bot-blocked at both onlinelibrary.wiley.com and the pdfdirect route; existence, authorship, OA status and abstract confirmed via api.crossref.org, api.openalex.org and doaj.org**
Delaney et al. (2020), Forecast Informed Reservoir Operations Using Ensemble Streamflow Predictions for a Multipurpose Reservoir in Northern California, Water Resources Research 56(9) September · CC BY 4.0 https://doi.org/10.1029/2019WR026604	Peer-reviewed open-access article, doi 10.1029/2019WR026604	Metadata confirmed via Crossref and OpenAlex: Water Resources Research vol. 56 issue 9, September 2020, AGU, CC BY 4.0, authors Delaney C; Hartman R; Mendoza J; Dettinger M; Delle Monache L;…	The operational algorithm actually deployed at Lake Mendocino, and the concrete…	**403-bot-blocked at agupubs.onlinelibrary.wiley.com; existence and metadata confirmed via Crossref/OpenAlex**

Source	Kind	What it provides	Relevance	HTTP
Brodeur, Delaney, Whitin & Steinschneider (2024), Synthetic Forecast Ensembles for Evaluating Forecast Informed Reservoir Operations, Water Resources Research 60(2) February 2 · CC BY 4.0 https://doi.org/10.1029/2023WR034898	Peer-reviewed open-access article, doi 10.1029/2023WR034898	Metadata confirmed via Crossref and OpenAlex: WRR vol. 60 issue 2, February 2024, AGU, CC BY 4.0, four authors exactly as named. OpenAlex reports a working OA PDF at…	Addresses AquaSync's evaluation problem head-on: you cannot honestly evaluate a…	**403-bot-blocked at agupubs.onlinelibrary.wiley.com; existence and metadata confirmed via Crossref/OpenAlex**
zpb4/syn-forecast_stochastic - synthetic hydrology and forecast generation with a FIRO model 1★ · pushed_at · MIT https://github.com/zpb4/syn-forecast_stochastic	GitHub repository (Python)	GitHub API confirms the repo exists: full_name zpb4/syn-forecast_stochastic, language Python, license MIT, archived false, created 2026-03-19, pushed 2026-08-19 (one week ago), stargazers_count 1,…	Live successor code to the 2024 synthetic-ensemble paper; the closest thing to a…	**200**
Albano et al. (2026), Characterizing Watershed Responses and Reservoir Operational Flexibilities: Analyses to Support Forecast-Informed Reservoir Operations (FIRO) Planning and Assessments, Water Resources Research 62(7) July 2026 · CC BY-NC 4.0 https://doi.org/10.1029/2025WR042131	Peer-reviewed open-access article, doi 10.1029/2025WR042131	Metadata confirmed via Crossref and OpenAlex: Water Resources Research vol. 62 issue 7, issued July 2026, AGU, CC BY-NC 4.0. Full author list Albano C; Shearer E; Forbis J; Yeates E; Kalansky J;…	A screening method for whether FIRO is viable at a given reservoir before committing to…	**403-bot-blocked (doi.org redirects to agupubs/Wiley, which blocks curl and WebFetch); existence and metadata confirmed via Crossref/OpenAlex**
F. M. Ralph, FIRO briefing to the California Water Commission, 16 April 2025 16 April 2 · California Water Commission public meeting material https://cwc.ca.gov/-/media/CWC-Website/Files/Documents/20…	Public-agency briefing slide deck (PDF, 35 pages, 4.7 MB)	Downloaded and parsed. Cover confirmed: F. Martin Ralph, Director CW3E at UCSD/SIO, FIRO National Expansion Pathfinder Chief Scientist. Pilot roster verified verbatim: Howard Hanson Dam (Green…	Authoritative snapshot of FIRO deployment status as of April 2025 - which sites have…	**200**
GAO-16-685: Army Corps of Engineers - Additional Steps Needed for Review and Revision of Water Control Manuals Published · US Government work - public domain https://www.gao.gov/products/gao-16-685	US Government Accountability Office audit report, published 26 July 2016	Retrieved via WebFetch (curl is hard-blocked). CONFIRMED: the Corps operates 'more than 700 dams and their associated reservoirs across the country'; engineer regulations mandate that water control…	Documents the institutional failure mode in the world's largest dam-operating agency:…	**403-bot-blocked to curl (both /products/ and /assets/gao-16-685.pdf); full content retrieved and verified via WebFetch**

Source	Kind	What it provides	Relevance	HTTP
Delft-FEWS FAQ (Deltares) Live; site · Zero-cost licence, sign the agreement for the operational build; source not public https://oss.deltares.nl/web/delft-fews/faq	Vendor FAQ page	Fetched and parsed. LICENSING CONFIRMED VERBATIM: 'Is Delft-FEWS open source? Delft-FEWS is Open Software, but not Open Source. Delft-FEWS is used in many critical operational systems, and our users…	The de facto operational forecasting platform AquaSync would have to sit beside or…	200
UW-SASWE/RAT - Reservoir Assessment Tool 3.0 26★ · pushed_at · GPL-3.0 - a real consideration if any of it is reused inside AquaSync https://github.com/UW-SASWE/RAT	GitHub repository (Python)	GitHub API confirms: full_name UW-SASWE/RAT, description 'Reservoir Assessment Tool', homepage http://satellitedams.net, language Python, license GPL-3.0, archived false, created 2022-05-12, pushed…	The satellite-based reservoir monitoring framework behind both Kerala papers, and the…	200
SASWE operational Kerala reservoir monitoring instance (RAT-WQ2 3.0) Live, but · Not stated on the page https://depts.washington.edu/saswe/kerala/	Liveweb dashboard (Plotly/Leaflet front end)	Fetched and parsed the page and its About sub-page. Confirmed: title 'RAT WQ2-Kerala', product name 'Reservoir Assessment Tool WQ2 (RAT-WQ2 3.0) - Kerala', tabs for Inflow, Outflow, Surface Area,…	A live Kerala deployment by the group that published the closest prior art. Worth…	200
Koo, Abraham, Jonoski & Solomatine (2025), Balancing Operators' Risk Averseness in Model Predictive Control for Real-time Reservoir Flood Control v2 29 Marc · arXiv preprint; journal version in Journal of Hydroinformatics https://arxiv.org/abs/2407.04506	arXiv preprint (eess.SY), v1 5 Jul 2024, v2 29 Mar 2025; published in Journal of Hydroinformatics 2025 (jh2025019), doi 10.2166/hydro.2025.019	Fetched and parsed the arXiv abstract page. DEPLOYMENT-FAILURE STATEMENT CONFIRMED VERBATIM: 'Despite many studies on reservoir flood control and their theoretical contribution, optimisation…	States AquaSync's central adoption risk in one sentence and prescribes the fix: an…	200
Rayner, Lach & Ingram (2005), Weather forecasts are for wimps: why water resource managers do not use climate forecasts, Climatic Change 69(2-3):197-227 Published · Subscription (Springer); Crossref lists only the Springer TDM licence https://link.springer.com/article/10.1007/s10584-005-3148…	Peer-reviewed article, doi 10.1007/s10584-005-3148-z	Landing page reachable (200) and metadata confirmed via Crossref: Climatic Change vol. 69 issue 2-3, pages 197-227, April 2005, Springer, authors Rayner S; Lach D; Ingram H. 277 citations per…	The organisational, not technical, explanation for why a working decision-support tool…	200
Google Flood Forecasting API Live docum · Data under CC BY 4.0, offered at no charge https://developers.google.com/flood-forecasting	Product/API documentation page	Fetched and parsed the served HTML. CONFIRMED from the page itself: 'The Flood Forecasting API provides access to real-time riverine flood forecasts for numerous countries, and is available at no… caveat: Access is waitlisted, not open. Budget lead time before assuming AquaSync can call this API.	The only operational, free, externally-maintained riverine forecast feed that could…	200

Verification notes

- BIGGEST CORRECTION - the 2018 data gap is only half-closed. CAG Appendices 3.5-3.7 do give 112 daily rows each, 10-06-2018 to 29-09-2018 (confirmed by date-sorted extraction, not string sort), but they are

SIMULATION tables. Cross-diffing Appendix 3.5 (1983 curve) against 3.6 (2020 curve) proves it: identical inflow, divergent level and spill. What AquaSync actually gains is a genuine daily 2018…

- THE RULE CURVE IS NOT INTERPOLATED, AND THIS MATTERS OPERATIONALLY. CAG Appendix 3.3 gives the 2020 curve at 18 ten-day steps and I verified all 36 values. The KSDMA bulletin of 25 Aug 2026 lists Idukki rule level 2390.09 ft = 728.4994 m, which is exactly the 31 August row - not the linear interpolation between 20 Aug (727.50) and 31 Aug (728.50), which would give 727.955 m. KSEB therefore…
- THE CENTRAL COMPLICATION SURVIVES SCRUTINY AND IS CORRECTLY STATED. I read Tables 3.6, 3.7 and 3.8 directly. Idukki actual spills 14-18 Aug 467.51 MCM; 1983 curve 558.19; 2020 curve 531.03; only the 1983 curve initialised from the LOWER rule level gives 308.13 MCM. Idamalayar 2020 curve 323.49 vs 340.50 actual (5% better). The 2020 rule curve KSEB actually adopted would have made Idukki worse.…
- THE FORECAST-SKILL FLOOR CLAIM IS THE STRONGEST FINDING IN THE SWEEP AND IT IS VERBATIM CORRECT. Current Science 116(5) p.789: 'The probability of detection (POD) of higher rainfall (>25 mm/day) at short range (3-day) or medium range (5-day) even with the ensemble of models is still low at about 20% to 30% only with a large probability of false alarm.' Set against Lake Mendocino's verified CNRFC…
- DATE AND VERSION DRIFT ON THE LIVE FEED. The KSDMA portal now shows 26/08/2026 while the KSEB PDF at the linked URL still holds the 25/08/2026 table (Last-Modified 25 Aug 12:08 GMT) - the page heading updates ahead of the file. More importantly, KSEB-SITE-13.pdf is overwritten in place and the numeric index rotates, so there is no archive at that URL. Any AquaSync ingestion must scrape the…
- FIVE 403s WERE GENUINE BOT BLOCKING, NOT DEAD LINKS, AND ALL FIVE RESOURCES ARE REAL - but note one important asymmetry. ascelibrary (Das et al.), Wiley/agupubs (Delaney, Brodeur, Albano, Forbis & Ly) and gao.gov all block automated agents; every one was confirmed via Crossref, OpenAlex, DOAJ or WebFetch. Delaney, Brodeur, Albano and Forbis & Ly are all open access (CC BY or CC BY-NC) and…
- CITATION COUNTS ARE SOURCE-DEPENDENT AND ALREADY STALE. Delaney et al. 2020: 114 per Crossref but 143 per OpenAlex. Brodeur et al. 2024: 18 per Crossref, 16 per OpenAlex. Forbis & Ly: 6-7. Albano et al.: 0. Rayner et al. 2005: 277 - by an order of magnitude the most-cited item here, which is itself the finding: the paper about why operators ignore forecasts is cited far more than any paper about…
- TWO REPOS, VERY DIFFERENT MATURITY - the original framing flattened this. UW-SASWE/RAT is a real project: 26 stars, 6 forks, GPL-3.0, created 2022, pushed 2026-01-13, with the three claimed releases confirmed to the day (v3.0.17 2025-04-11, v3.0.18 2025-08-18, v3.0.19 2025-11-25 'with SWOT Plugin'). zpb4/syn-forecast_stochastic has 1 star, 0 forks, was created 2026-03-19, and contains only…
- THE STRONGEST ACTIONABLE FINDING IS THE CHEAPEST TO IMPLEMENT, AND IT IS NOT ABOUT FORECASTS. Koo et al., verified verbatim: 'optimisation methodologies have not been widely applied in real-time operation due to disparities between research assumptions and practical requirements. To address this gap, we include practical objectives, such as minimising the magnitude and frequency of changes in…
- AN UNEXPECTED LEAD ON THE PRE-2020 DATA GAP, found while checking a different claim. The Google Flood Forecasting page itself points to the Google Runoff Reanalysis & Reforecast dataset (GRRR), described as global river discharge estimates covering 1980-2023, and an Inundation History dataset covering 1999-2020 - both separate from the waitlisted forecast API and neither in the source list.…

Rejected (9)

Recorded so the same ground is not covered twice.

Claimed source	Why it was rejected
CLAIM (not the source): CAG Appendices 3.5-3.7 are 'genuine 2018 Idukki and Idamalayar daily operational data' including reservoir level and spills	REJECTED AS DESCRIBED - the single most consequential error in the sweep. The appendices are titled 'Sample simulations of Idukki/Idamalayar reservoir using the 1983/2020 rule curve'. I diffed Appendix 3.5 against 3.6…
CLAIM: Vegad & Mishra project days above 90% storage rising 'from ~6 at 1C to 23 at 3C'	REJECTED - the numbers are wrong. I read the OA PDF. The paper states 'a threefold increase in the median number of days, from 4.7 +/- 6.7 (median +/- 1 std) days at 1.0 C warming to 14.6 +/- 15.11 days at 3.0 C…
CLAIM: the 19 Sep 2020 SANDRP post contains 'the IIT Madras team's defence AND Harsha's reply... the only place the full argument appears in one thread'	REJECTED as described - that page carries ONLY the Sudheer et al. response. Harsha's rejoinder is a separate post published the same day, which I located from its 'Next Post' link and verified 200:…
Delft-FEWS About page deployment statistics (40+ operational forecasting centres since 2002/2003, 100+ information sources, 50+ simulation modules, HEC/MIKE/HBV/SOBEK)	REJECTED - I fetched the page (200, 141 KB) and searched the rendered text for 'forty', '100', '50', 'SOBEK', 'HBV', 'MIKE', '2002', '2003', 'forecasting centres' and 'operational'. None occur. What is served at that…
CLAIM: Google Flood Forecasting coverage statistics (7-day horizon, 5,000+ verified gauges, ~250,000 virtual gauges, 150+ countries, Mean Embedding Forecast LSTM as of Dec 2025, all-India CWC partnership with 1,000+ hourly gauges)	REJECTED as sourced - I fetched and searched the served page for every one of those strings and none appear. The page confirms only that the API provides real-time riverine flood forecasts for numerous countries, free…
CLAIM: SASWE Kerala instance is 'funded by NASA, KSCSTE and CWRDM'	REJECTED - not verifiable from the site. I fetched both the main page and about.html and searched for 'NASA', 'KSCSTE', 'CWRDM', 'fund' and 'acknowledg'. None occur. The page attributes development to Sarath Suresh at…
CLAIM: the HESS discussion thread comprises '1 referee report, 3 short comments, 5 author responses'	REJECTED on the referee count - the live thread lists TWO referee comments (RC1 by Sudheer Padikkal, 27 Sep 2018; RC2 'Comments on Mishra and others' by Amit Kumar, 27 Oct 2018), three short comments (SC1 Jeroen van…
gao.gov direct PDF asset	REJECTED as a fetch route - returns 403 to curl with a browser user agent, as does the /products/ page. Use the product page URL (kept in the confirmed list) and open it in a browser, or via WebFetch, which succeeded…
CLAIM: 'Forbis et al. (2025)' with specific figures - more than 400 WCMs, 'would take decades', 'nearly exclusively based on rule curves', FIRO Screening Process steering committee 2021, SPD beta test 2022-2024, Stage A/Stage B	PARTIALLY REJECTED - two problems. (1) Attribution: the paper has exactly two authors, Joe Forbis (USACE ERDC-CHL) and Cuong Ly (USACE SPD), so it is 'Forbis & Ly', not 'Forbis et al.'. (2) All the specific figures…

9 · Integrating With Existing Dam, Scada And Government Water Systems (Periyar / Idukki / Idamalayar, Aquasync)

Confirmed sources (30)

Source	Kind	What it provides	Relevance	HTTP
KSEB Dam Safety Organisation — Monthly Statistics (download listing) n/a★ · Latest wor · Government of Kerala / KSEB Ltd — no explicit open licence stated https://dams.kseb.in/?p=329	Government data archive (WP Download Manager listing)	Page title confirmed: 'Monthly Statistics – KSEB Limted Dam Safety Organisation'. I parsed the HTML and counted exactly 91 unique wpdmdl download IDs, min 310, max 6380, with 66 occurrences of the… caveat: 91 are LISTED but only 86 are DOWNLOADABLE. IDs 310-314 (Jan-May 2019) return HTTP 302 to…	Authoritative KSEB copy covering the entire 2020-09-25 to 2021-04-30 window that is…	200
KSEB monthly workbook, June 2019 — the 2018 previous-year column (worked proof) n/a★ · Static arc · Government of Kerala / KSEB Ltd https://dams.kseb.in/?wpdmdl=315	Excel workbook (.xlsx), government primary record	Downloaded and parsed with openpyxl. Content-disposition filename 'June 2019.xlsx', Content-Length 1,511,065 — byte-exact to the claim. 31 sheets: 'Sheet1 (2)' plus '01.06.19 ' through '30.06.19'.… caveat: The leftover 'Sheet1 (2)' is titled 'STATUS OF WATER LEVEL (25.09.2018)' — a full genuine 2018 KSEB status sheet (IDAMALAYAR 160.84 m,…	Directly attacks KNOWN GAP 1 (no genuine pre-2020 Idukki/Idamalayar data). The 'Previous…	200
KSEB 2019 workbook set (Jun-Oct) — the full 2018 back-series, independently recounted n/a★ · Static arc · Government of Kerala / KSEB Ltd https://dams.kseb.in/?wpdmdl=500	Excel workbooks (.xlsx) — July 2019; sibling IDs 501/502/503 = Aug/Sep/Oct 2019	I downloaded June, Jul, Aug, Sep, Oct, Nov and Dec 2019 and programmatically counted every sheet that yields an Idukki 'Previous Year Water Level' value: Jun=31, Jul=32, Aug=10, Sep=9, Oct=22, total… caveat: 104 rows collapse to only 96 UNIQUE 2018 dates — the Aug/Sep 2019 workbooks are 3-hourly flood sheets that repeat the same previous-year…	The concrete GAP-1 deliverable: roughly 100 days of genuine 2018 daily Idukki AND…	200
KSEB monthly workbook, July 2026 — modern daily schema with rule curve and alert bands n/a★ · July 2026, · Government of Kerala / KSEB Ltd https://dams.kseb.in/?wpdmdl=6380	Excel workbook (.xlsx), government primary record	Content-disposition filename '07. July.xlsx', 252,222 bytes, 33 sheets (31 daily plus 'MU' and 'FB'). Daily-sheet header verified verbatim and RICHER than previously reported: Sl.No | Name of… caveat: The '% Storage wrt Live Storage' column contains #REF! formula errors in the cached values — recompute it rather than reading it. The 'MU'…	The single most operationally important find for the policy search. AquaSync currently…	200
KSEB Ltd live dam water level dashboard n/a★ · Live; as-o · Government of Kerala / KSEB Ltd https://kseb.in/dashboard/view/waterlevel	Government HTML dashboard with a date query parameter	Parsed the table: exactly 17 columns, header verbatim as claimed (Name of Dam/Reservoir, District, Today's Water level (metre), Today's Live Storage (MCM), %Storage wrt Live Storage at FRL, Same day… caveat: Archive is genuinely short: ?date=2026-07-01 and ?date=2025-10-15 both return the empty shell. Also the Blue/Orange/Red columns did NOT…	Free live feed of the alert bands the policy search must respect, plus the 1459.49 MCM…	200

Source	Kind	What it provides	Relevance	HTTP
Kerala SLDC — System Statistics: Storage (daily observed inflow, WITH a working date archive) n/a★ · Live daily · Government of Kerala / KSEB SLDC — no explicit licence https://sldckerala.com/index.php?id=7	Government HTML table, POST-queryable by date	Current-day row verified exactly as claimed for 25.08.2026: IDUKKI MDDL 694.944 | FRL 732.43 | Eff. Storage at FRL 1460 mcm | Gen. Cap at FRL 2190 mu | Level 720.968 | Storage 869.358 mcm | 60% |… caveat: Archive starts in early August 2019: 2019-08-05 returns an empty page, 2019-08-08 returns data. Verified working at…	The answer to KNOWN GAP 3 (perfect foresight). A daily OBSERVED inflow series in mcm/day…	200
KSDMA — Dam Water Level (daily consolidated KSEB + Irrigation bulletins) n/a★ · Updated da · Government of Kerala / KSDMA https://sdma.kerala.gov.in/dam-water-level/	Government daily bulletin, PDF	Confirmed: two daily PDFs, 'Water Levels of Major Reservoirs (KSEB)' and '(IRRIGATION)', both stamped 26/08/2026 11:00 AM, plus a 'Guidelines to interpret the tables of KSEB and IRRIGATION' PDF. I… caveat: IMPORTANT INCONSISTENCY: for the same date the KSDMA bulletin and the KSEB dashboard disagree. KSDMA (25/08/2026) gives Idukki WL 2364.22…	The consolidated cross-agency bulletin, and the only source that publishes the…	200
KSDMA — Dam Management (the GO 453/2021/DMD governance trigger) n/a★ · Current li · Government of Kerala / KSDMA https://sdma.kerala.gov.in/dam-management/	Government policy page with EAPand rule-curve index	I ADDED this source — the previous report asserted the GO as a key finding with NO url at all. Verified verbatim on the page: 'The State Government has constituted a multi-agency committee for… caveat: The KSEB rule-curve and EAP links are largely MISSING: of the 19 KSEB dams only 'Kallarkutty' carries a link, the Idukki-Cheruthoni EAP…	This is the decision point AquaSync should target. It confirms the intervention is a…	200
CWC — Study Report: Kerala Floods of August 2018 (Hydrological Studies Organisation, Sept 2018) n/a★ · Static; ho · Government of India / CWC — public report https://sdma.kerala.gov.in/wp-content/uploads/2020/08/CWC…	Government technical report, PDF (text-extractable)	Downloaded: 6,003,314 bytes, 48 pages, 72,025 extractable characters. Page 1 confirms 'Government of India / Central Water Commission / Hydrological Studies Organisation / STUDY REPORT KERALA FLOODS… caveat: The hourly detail is a NARRATIVE plus Fig.4 (a chart image). There is no tabulated hourly inflow/outflow/level series in the text — you…	The only verified source for the 9-17 Aug 2018 peak that the KSEB workbooks do not…	200
CWC — Report on Kerala Flood & Solutions (December 2018) n/a★ · Static; ho · Government of India / CWC — public report https://sdma.kerala.gov.in/wp-content/uploads/2020/10/Ker…	Government technical report, PDF (text-extractable)	Downloaded: 9,744,067 bytes, 60 pages, 78,752 extractable characters. Section 8.3.2.2 verified verbatim: 'It is proposed to issue level flood forecast for Neeleswaram using dam discharges of the… caveat: The report contradicts itself on the Neeleswaram intervening free catchment area: 2,329 km2 in one sentence and 3,042 km2 two sentences…	Two things AquaSync needs. (1) GAP 2: an official government-published 8-hour lag for…	200

Source	Kind	What it provides	Relevance	HTTP
Encardio Rite — Idukki Dam monitoring case study (the incumbent SHM/EWS) n/a★ · Undated ve · Vendor copyright — reference only https://www.encardio.com/blog/idukki-dam-monitoring	Vendor case study / blog	Fetched and checked line by line. CONFIRMED on the page: the contract name 'Real-Time Early Warning Structure Health Monitoring and Data Interpretation (RESHMI)'; that it is 'a part of the Dam… caveat: The earlier report presented the Pallam/Vazhathoppe server locations as 'Verified via' this page. They are not on it. Drop that detail or…	Establishes that Idukki's real-time instrumentation is an existing, closed,…	200
NDSA public document API (undocumented but open) n/a★ · PDFs serve · Government of India / NDSA https://ndsa.gov.in/ndsa-backend/web/api/document-config	Open JSON API (no key)	GET returns exactly the claimed document-type map: guidelines-manual-code/Guidelines/guidelines, reports/Reports/reports, order-notices/OrderAndNotice/order-and-notice, … caveat: Two regulations the earlier report missed are equally relevant to a telemetry layer: Reg 10 (24/04/2024, data requirements of…	Turns the compliance hook into something machine-checkable — you can poll for newly…	200
NDSA Reg 9 — forwarding the analysis of instrument readings to the SDSO (s.32(2)) n/a★ · Gazetted 2 · Government of India — Official Gazette https://ndsa.gov.in/ndsa-backend/web/sites/default/files/…	Gazette notification, PDF (scanned image)	200, Content-Type application/pdf, 1,906,475 bytes, 2 pages. Title and 24/04/2025 publish date confirmed against the NDSA API listing. caveat: CONFIRMED SCANNED: pypdf extracts 0 characters across 2 pages. The actual reporting cadence and form must be read by eye or OCR'd. Do not…	The statutory basis for making AquaSync emit an SDSO-facing instrument-reading report —…	200
NDSA Reg 8 — minimum instrumentation in specified dams (s.32(1)) n/a★ · Gazetted 2 · Government of India — Official Gazette https://ndsa.gov.in/ndsa-backend/web/sites/default/files/…	Gazette notification, PDF (scanned image)	200, application/pdf, 1,725,803 bytes, 2 pages. Title and 24/04/2024 date confirmed against the NDSA API listing. caveat: pypdf extracts 0 characters across 2 pages — scanned.	Defines the minimum instrument set a specified dam must carry — the floor of what…	200
NDSA Reg 16 — data requirements of hydro-meteorological stations near specified dams (s.33(1)) n/a★ · Gazetted 2 · Government of India — Official Gazette https://ndsa.gov.in/ndsa-backend/web/sites/default/files/…	Gazette notification, PDF (scanned image)	200, application/pdf, 328,230 bytes, 1 page. Title and 20/05/2024 date confirmed against the NDSA API listing. caveat: pypdf extracts 0 characters from the single page — scanned.	Prescribes what hydro-met data a dam owner must collect — the statutory shape of the…	200
The Dam Safety Act, 2021 (Act 41 of 2021) — full text n/a★ · Static sta · Government of India — statute, public https://wrd.maharashtra.gov.in/Site/Upload/Pdf/DSA_2021_a…	Statute, PDF (text-extractable)	459,259 bytes, 21 pages, 64,912 extractable characters. Page 1: 'THE DAM SAFETY ACT, 2021 / NO. 41 OF 2021 / [13th December, 2021.]'. I verified the rule-making clause list against the regulation… caveat: This is a Maharashtra WRD-hosted copy, not an india-code.nic.in original — content verified but the host is a state department. Note the…	The parent statute every NDSA regulation hangs off. It is what makes the telemetry and…	200
PIB — 'Dam Rehabilitation: Strengthening Infrastructure through Policy and Technology' (15 May 2026) n/a★ · Published · Government of India / PIB — public https://static.pib.gov.in/WriteReadData/specificdocs/docu…	Government press-information backgrounder, PDF (text-extractable)	569,643 bytes, 8 pages, 15,939 extractable characters, dated 'May 15, 2026'. I grep-verified every claimed figure: India 3rd worldwide with 6628 specified dams, 6,545 operational and 83 under… caveat: None material.	Kerala's inclusion in DRIP Phase I is what puts Idukki's RESHMI system inside a…	200

Source	Kind	What it provides	Relevance	HTTP
NDSA Rapid Risk Screening / DHARMA dashboard API n/a★ · 2026 pre-m · Government of India / NDSA https://rrssd.ndsa.gov.in/api/dharma/dashboard	Open JSON API (no key)	Returned byte-for-byte what was claimed:… caveat: Undocumented endpoint — it can change or disappear without notice. It reports inspection counts only; it is not a dam-data API.	Evidence that NDSA already exposes open JSON endpoints — a useful precedent when arguing…	200
CERT-In Directions under s.70B(6) of the IT Act 2000, dated 28 April 2022 n/a★ · In force s · Government of India — public direction https://www.cert-in.org.in/PDF/CERT-In_Directions_70B_28.…	Binding government direction, PDF (text-extractable)	434,961 bytes, 8 pages, 12,899 extractable characters. Header confirmed: 'No. 20(3)/2022-CERT-In / MeitY / Indian Computer Emergency Response Team / Dated: 28 April, 2022 / Subject: Directions under… caveat: None material.	Hard constraints on where AquaSync's logs live and how fast an incident must be…	200
CEA (Cyber Security in Power Sector) Guidelines, 2021 — authoritative CEA copy n/a★ · Issued 202 · Government of India / CEA — public https://cea.nic.in/wp-content/uploads/notification/2021/1…	Government sectoral guidelines, PDF (text-extractable)	I REPLACED the previously reported aegcl.co.in mirror with this CEA-hosted original and proved they are the same file (both md5 40498ff8ccad5a6d56a020018a1a84d8, 567,765 bytes). 24 pages, 55,202… caveat: Two specific compliance numbers previously attributed to this document are not in it. Quote the cardinal-principle isolation language…	The actual quotable basis for a read-only architecture. The 'hard isolation of OT from…	200
CWC — Flood Forecasting / Hydrological Observation n/a★ · Current li · Government of India / CWC https://cwc.gov.in/flood-forecasting-hydrological-observa…	Government programme page	Text scraped and grepped. CONFIRMED verbatim: 'Flood Forecasting Network covers 325 stations covering 197 low lying area/ cities and towns besides 128 reservoirs all over the country. The network is… caveat: Drop the 878 and 1,054 figures unless someone sources them separately. Weigh this page against the CWC Dec-2018 report's own statement…	Establishes that CWC already issues inflow forecasts to reservoir operators at 128…	200
CWC Advisory Flood Forecast (AFF) — methodology n/a★ · Live servi · Government of India / CWC https://aff.india-water.gov.in/method.php	Government model documentation page	Fetched and confirmed on every point: 'CWC is currently doing rainfall-based mathematical modelling for flood forecasting which uses both the hydrologic (rainfall-runoff) and hydrodynamic modelling… caveat: I could NOT confirm AFF covers the Periyar or Neeleswaram: the landing HTML contains no occurrence of 'Neeleswaram', 'Periyar' or 'Kerala'…	The most direct answer to GAP 3. A 7-day, 3-hourly national inflow-forecast product…	200
OGC SensorThings API Part 1: Sensing, Version 1.1 (OGC 18-088) n/a★ · Published · OGC document licence — free to read and implement https://docs.ogc.org/is/18-088/18-088.html	Open standard, Implementation Standard	HTML title confirmed exactly: 'OGC SensorThings API Part 1: Sensing Version 1.1'. caveat: India has not adopted SensorThings — this is a design choice AquaSync would be making unilaterally, not a compliance requirement.	The interoperability target if a partner agency (KSDMA, NDSA, a DRIP consultant) asks…	200

Source	Kind	What it provides	Relevance	HTTP
OGC API - Environmental Data Retrieval Standard (OGC 19-086r6) n/a★ · Published · OGC document licence https://docs.ogc.org/is/19-086r6/19-086r6.html	Open standard, Implementation Standard	HTML title confirmed exactly: 'OGC API - Environmental Data Retrieval Standard'. Companion document OGC WaterML 2 Part 3 / HY_Features (https://docs.ogc.org/is/14-111r6/14-111r6.html) also verified… caveat: A voluntary choice, not a mandate.	The lighter-weight fit for AquaSync: EDR's position/radius/area/cube/trajectory/corridor…	200
pygeoapi 623★ · pushed 202 · MIT https://github.com/geopython/pygeoapi	GitHub repository (Python server)	GitHub API confirms the repo exists: 623 stars, pushed 2026-08-18T22:37:18Z, licence MIT, archived=false, fork=false. Description confirms it is 'a Python server implementation of the OGC API suite… caveat: The >=3.12 floor is a real constraint if AquaSync is pinned to an older interpreter.	The pragmatic path to a standards-compliant read API over AquaSync's existing Python…	200
FROST-Server (Fraunhofer IOSB) — OGC SensorThings reference implementation 227★ · pushed 202 · LGPL-3.0 https://github.com/FraunhoferIOSB/FROST-Server	GitHub repository (Java server)	GitHub API: 227 stars, pushed 2026-08-24T06:36:31Z, licence LGPL-3.0, archived=false. Description: 'A Complete Server implementation of the OGC SensorThings API'. Docker Hub cross-check:… caveat: Java plus PostgreSQL/PostGIS plus Docker — a separate operational stack alongside AquaSync's Python. Treat as optional.	Only worth the operational cost if a partner agency demands SensorThings conformance;…	200
opcua-asyncio (asyncua) 1468★ · pushed 202 · LGPL-3.0 https://github.com/FreeOpcUa/opcua-asyncio	GitHub repository (Python protocol library)	GitHub API: 1,468 stars, pushed 2026-08-24T12:47:48Z, licence LGPL-3.0, archived=false, description 'OPC UA library for python >= 3.10'. caveat: Speculative until an actual OPC-UA endpoint is confirmed to exist at Kalamassery or Pallam. No such endpoint is currently evidenced.	The most plausible of the protocol clients: if KSEB ever exposes a read-only DMZ…	200
iec104-python / c104 (Fraunhofer FIT) — IEC 60870-5-104 in Python 101★ · pushed 202 · GPL-3.0 https://github.com/Fraunhofer-FIT-DIEN/iec104-python	GitHub repository (Python binding over lib60870-C)	GitHub API: 101 stars, pushed 2026-08-06T13:35:46Z, licence GPL-3.0, archived=false. Description: 'A Python module to simulate SCADA and RTU communication over protocol 60870-5-104 to research ICT… caveat: GPL-3.0 — a copyleft obligation if AquaSync ever links it into a distributed product.	IEC 60870-5-104 is the protocol the CEA guidelines name for RTU/PLC conformance, so this…	200
libiec61850 (MZ Automation) — IEC 61850 incl. the -7-410 hydro profile substrate 1212★ · pushed 202 · GPL-3.0 https://github.com/mz-automation/libiec61850	GitHub repository (C library)	GitHub API: 1,212 stars, pushed 2026-08-26T11:06:17Z (today), licence GPL-3.0, archived=false. Description: 'Official repository for libIEC61850, the open-source library for the IEC 61850… caveat: GPL-3.0 (a commercial licence is sold separately). The hydro logical-node model is in the paid IEC standard, not in this repo.	IEC 61850-7-410 defines the logical nodes for hydropower (gates, turbines, reservoirs);…	200

Source	Kind	What it provides	Relevance	HTTP
Eclipse Tahu — MQTT Sparkplug B reference implementations 264★ · pushed 202 · EPL-2.0 https://github.com/eclipse-tahu/tahu	GitHub repository (multi-language reference implementation)	GitHub API: 264 stars, pushed 2026-08-10T19:37:35Z, licence EPL-2.0, archived=false. Description confirms it 'addresses the existence of legacy SCADA/DCS/ICS protocols and infrastructures and… caveat: Sparkplug v3.0 is the current spec version; verify against the Eclipse Sparkplug specification repo rather than assuming Tahu tracks it…	If AquaSync ever publishes telemetry outward rather than reading inward, Sparkplug B's…	200

Verification notes

- BIGGEST CORRECTION — THE SLDC ARCHIVE IS REAL. The previous agent concluded 'I could NOT get the historical date form to work ... treat it as a forward-scraping daily snapshot, not an archive.' That is wrong, and it badly understates the single most useful source for GAP 3. The form works as a POST to https://sldckerala.com/index.php?id=7 with body…
- THE ALERT-BAND FINDING IS BETTER THAN CLAIMED AND NOW HAS AN EXACT RULE. The earlier report said the bands 'MOVE with the rule curve through the monsoon' based on comparing a July workbook against an August dashboard. I read four sheets inside the single July-2026 workbook and the bands move WITHIN the month: rule 2375.33 ft on 01.07 and 10.07, 2377.95 on 20.07, 2380.58 on 31.07. More usefully,…
- THE THREE OFFICIAL KERALA SOURCES DISAGREE WITH EACH OTHER ON THE SAME DATE. For 25/08/2026: the KSDMA daily KSEB bulletin gives Idukki water level 2364.22 ft with rule level 2390.09 ft and bands 2382.09/2388.09/2389.09; the KSEB dashboard gives 720.968 m (about 2365.38 ft) with bands 2389.78/2395.78/2396.78 (implied rule 2397.78); SLDC gives 720.968 m with remarks 2365.38 ft. SLDC and the…
- THE 2018 RECOVERY IS REAL BUT MUST BE STATED PRECISELY, AND THE FLOOD PEAK IS NOT IN IT. My independent parse reproduced the 104-row count exactly (Jun 31, Jul 32, Aug 10, Sep 9, Oct 22) and confirmed Nov/Dec 2019 dropped the column. But 104 rows collapse to only 96 unique 2018 dates, and only THREE of them are in August 2018 — the 20th, 30th and 31st. The 9-17 August 2018 flood peak, the whole…
- THE KSEB ARCHIVE IS 86 FILES, NOT 91, AND CHANGES FILE FORMAT MID-STREAM. 91 wpdmdl IDs are listed but IDs 310-314 (Jan-May 2019) 302-redirect to a dead kseboa.in host. Working range is 315 (June 2019) through 6380 (July 2026). Also, ID 1178 (Nov 2020) onward are legacy .xls, not .xlsx — openpyxl will refuse them, so a loader needs xlrd or a conversion step. This matters because the .xls stretch…
- TWO CEA COMPLIANCE NUMBERS IN THE EARLIER REPORT ARE NOT IN THE CEA DOCUMENT. The '6-month IT audit / annual OT audit' cadence and the 'training within 90 days of notification' deadline do not appear in the 55,202 characters of the CEA Guidelines 2021. The real quotable provisions are stronger for AquaSync's purposes anyway: the cardinal principle 'There is hard isolation of their OT Systems…
- THE GOVERNANCE TRIGGER HAD NO URL AND NOW DOES. G.O.(Rt) No. 453/2021/DMD was presented as a key finding with no source at all — precisely the failure mode the brief warns about. I located and verified it at https://sdma.kerala.gov.in/dam-management/ (200), where the text appears verbatim including 'The committee meets when dams reach their orange level of alert or if extreme rainfall events are…
- THE CWC 8-HOUR TRAVEL TIME IS REAL BUT IS A 2018-ONLY WORKING ASSUMPTION, NOT A CALIBRATED LAG. Verified verbatim in sec 8.3.2.2. But read the full sentence: the analysis 'was done only for 2018 using the outflow hydrograph from calibrated Mike-11 model at Idukki and Idamalayar dams' because 'hourly discharge from these project authorities were not available'. So the 8 h was derived against…
- THE SEPT-2018 CWC REPORT IS NARRATIVE PLUS CHARTS, NOT A DOWNLOADABLE HOURLY SERIES. Every one of the 26 numbers I checked is verbatim in the text — 2532 cumec at 22:00 on 15 Aug,

41% attenuation, FRL touched 01:00 on 17 Aug, Idamalayar 168.06 m / 1164 cumec / 1271 cumec average spill, Neeleswaram 8800 cumec at EL 12.4 m with about 7300 cumec from the free catchment, Periyar 4,033 km2, free…

- UNDER-REPORTED VALUE WORTH PROMOTING, AND ONE SOURCE DEMOTED. Three things the earlier sweep found but undersold: (1) the KSEB daily sheets carry Power House Discharge, Spill (MCM) and Average Inflow (Cumec) columns omitted from the schema description — observed release partitioning, directly useful for calibrating the hydropower and spill models; (2) the CWC Dec-2018 report contains…

Rejected (13)

Recorded so the same ground is not covered twice.

Claimed source	Why it was rejected
KSEB monthly workbooks for Jan-May 2019 (wpdmdl IDs 310-314)	Confirmed DEAD. Each of 310, 311, 312, 313, 314 returns HTTP 302 to http://kseboa.in/wp-content/uploads/2020/05/{January,February,March,April,May}-2019.xlsx, and kseboa.in does not resolve (curl 000, final code 302,…
CEA guidelines claim: 'IT systems audited at least every SIX MONTHS and OT systems at least ANNUALLY'	NOT IN THE DOCUMENT. I extracted all 55,202 characters and searched for 'audit', 'six months', 'half-yearly', 'half yearly', 'once in a year', 'every year' and 'annual'. There is no audit-cadence requirement of any…
CEA guidelines claim: 'mandatory CEA-designated cyber-security training for all IT/OT O&M; personnel within 90 days of notification'	MISREAD. All four '90 days' occurrences refer to retaining cyber logs and cyber forensic records for at least 90 days from commissioning or recovery — not to a training deadline. What the document DOES say (Article 8)…
CWC claim: '878 hydrological and hydro-meteorological sites' and 'about 1,054 telemetry-equipped stations', transmitted by 'wireless, telephone/mobile, VSAT and satellite telemetry'	NOT ON THE CITED PAGE. I scraped the full rendered text (8,996 characters) and grepped: '878' zero hits, '1054'/'1,054' zero hits, 'telemetr' zero hits, 'VSAT' zero hits, 'satellite' zero hits. The page's own…
Encardio claim: Drishti servers at KSEB Pallam (Kottayam) and Vazhathoppe (Idukki); World Bank attribution on this page	NOT ON THE CITED PAGE. A full fetch confirms RESHMI, DRIP, two robotic total stations, the sensor list, BusMux/GSM-GPRS and Drishti — but 'Pallam', 'Kottayam', 'Vazhathoppe' and 'World Bank' are all absent. The earlier…
KSEB/SLDC protocol-boundary claim: 73 telemetered stations, 50 SCADA-enabled substations integrated to SLDC Kalamassery over ICCP (IEC 60870-6 TASE.2), distribution SCADA/DMS in Thiruvananthapuram/Ernakulam/Kozhikode, SCADA/EMS upgrade under administrative sanction	UNSOURCED AND UNCONFIRMED. The earlier report presented this as a key finding with NO url whatsoever. I found the nearest candidate page (KSEB Transmission, HTTP 200) and fetched it: it confirms ONLY that 'The System…
pymodbus (verified real, but dropped from the integration list)	THE REPO IS REAL — GitHub API confirms 2,762 stars, pushed 2026-08-25T21:23:53Z, not archived, licence reported as NOASSERTION (GitHub cannot map its LICENSE file to an SPDX id). I am dropping it from the confirmed…
India-WRIS / NHP RTDAS plus ArcGIS-server claim (arc.indiawris.gov.in/server/rest/services/...)	UNREACHABLE AND UNVERIFIED. No URL was given in the source list, only in a key finding. I tested https://indiawris.gov.in/wris/, https://arc.indiawris.gov.in/server/rest/services and the ?f=json form — all three return…
NCIIPC guideline documents / nciipc.gov.in	UNREACHABLE, as the earlier report honestly flagged. I went further: DNS DOES resolve (nciipc.gov.in to 164.100.59.137) but the TCP connection to port 443 times out after 20 s, and both http:// and https:// return curl…
CEA guidelines hosted on aegcl.co.in (third-party mirror)	SUPERSEDED, not wrong. This mirror returns 200 and is byte-identical to the CEA original (both md5 40498ff8ccad5a6d56a020018a1a84d8, 567,765 bytes) — but it is hosted by Assam Electricity Grid Corporation, not the…
NDSA regulations human landing page (as a data source)	RETURNS 200 BUT SERVES NO DATA. It is a JavaScript SPA: the server response is a 1,862-byte shell with 36 characters of visible text ('Home | National Dam Safety Authority'). None of 'Reg 9', 'Reg 8', 'Reg 16', 'log…

Claimed source	Why it was rejected
'OGC HydroDART'	CONFIRMED NON-EXISTENT — the earlier agent's rejection was correct and I restate it so it does not creep back in. A targeted search for 'HydroDART' plus OGC standard returns nothing at OGC or anywhere else; results are…
IDRB Kerala 'Dam Safety Rule Curve' page	DOES NOT RESOLVE. Surfaced in a search as a promising route to Kerala's official rule curves, but curl returns 000 (no connection). Not reportable. The KSDMA Dam Management page is the working substitute, though its…

What is now on disk

Everything below was downloaded into `research/` and verified non-empty. Bulk snapshots are gitignored - they are large and re-fetchable - while the manifest, index and findings are committed so the research is reproducible.

Repository snapshots are extracted from GitHub tarballs rather than cloned. Two of them ship filenames containing colons, which are illegal on Windows and abort a git checkout entirely; tar skips those entries and extracts the remaining files.

Acquired (14)

Resource	Kind	Local path	Size	Why it is here
pastas	repo	`research/repos/pastas`	164.0 MB	Hydrological time-series analysis. Large; low direct relevance.
google-flood-forecasting	repo	`research/repos/google-flood-forecasting`	140.1 MB	Google's open-sourced hydrology framework behind Flood Hub.
NOAA-t-route	repo	`research/repos/t-route`	81.7 MB	Operational Muskingum-Cunge routing used in the US National Water Model.
Sentinel1-Flood-Finder	repo	`research/repos/Sentinel1-Flood-Finder`	42.9 MB	SAR flood extent mapping. Validates extent, never timing (6-12 day revisit).
opendatakerala-lsg-boundaries	dataset	`research/repos/lsg-kerala-data`	30.2 MB	Kerala ward/panchayat boundaries as GeoJSON. Needed for downstream exposure mapping.
AI4Water	repo	`research/repos/AI4Water`	26.3 MB	ML framework for hydrological time series. MIT.
LFPtools	repo	`research/repos/LFPtools`	24.0 MB	LISFLOOD-FP preprocessing. Last push 2021.
neuralhydrology	repo	`research/repos/neuralhydrology`	22.9 MB	LSTM rainfall-runoff, the reference implementation. Directly relevant to the forecast-error gap.
Kerala-Dam-Water-Levels	dataset	`research/repos/Kerala-Dam-Water-Levels`	15.4 MB	The KSEB bulletin feed AquaSync already uses. 18 KSEB dams + irrigation. Verified: covers 2020-08 to 2026-08, NOT 2018.
rtc-tools	repo	`research/repos/rtc-tools`	6.5 MB	Deltares reservoir optimisation framework. LGPL-3.0, active.
pywr	repo	`research/repos/pywr`	4.3 MB	Water resource network allocation model. GPL-3.0.
pyflwdir	repo	`research/repos/pyflwdir`	2.4 MB	Deltares flow-direction / river-network derivation from a DEM. MIT.
dMC-Juniata-hydroDL2	repo	`research/repos/dMC-Juniata`	159.6 KB	Differentiable Muskingum-Cunge. Small; directly relevant to routing calibration.
idukki-live-bulletin	dataset	`research/sources/datasets/kerala_dams_live.json`	16.9 KB	Today's bulletin snapshot for all dams.

Refresh everything

```
python scripts/acquire.py # fetch the whole manifest
python scripts/acquire.py --priority high # just the important ones
python scripts/acquire.py --index-only # rebuild the index only
python scripts/build_research_report.py # rebuild this report
```

Method

Thirteen agents in three phases. Six searched one domain each; six adversarial verifiers re-checked those findings; one synthesised.

Phase	Agents	What they did
Research	6	One per domain. Many distinct web searches, primary sources preferred, every URL curl-verified with the real HTTP status recorded, GitHub repos checked against the API for stars, licence and last push.
Verify	6	Adversarial. Re-checked every URL independently, deep-read the most important sources to confirm they provide what was claimed, and rejected anything unconfirmable. Bot-blocked sites (403/503 to automated requests) were confirmed by another route rather than discarded.
Synthesize	1	Cross-domain integration paths, prioritised upgrades tied to specific verified projects, a download manifest, and the evidence against the thesis.

Domains covered

Domain	Confirmed	Rejected
Historical Kerala / India reservoir and flood data, especially pre-2020	30	1
Open-source reservoir operation, optimisation and forecast-informed operations (FIRO)	24	5
Open-source river routing, hydraulics and 2D flood inundation (adversarial verification pass)	30	2
State of the art in dam operation decision support: what is deployed, what fails	30	9
Integrating with existing dam, SCADA and government water systems (Periyar / Idukki / Idamalayar, AquaSync)	30	13

What this method does not guarantee.

Verification confirms that a URL resolves and that its content matches what was claimed about it. It does not constitute a peer review of the underlying science, and it cannot detect a source that is internally consistent but wrong. Anything load-bearing for a decision should still be read in full before it is relied on.